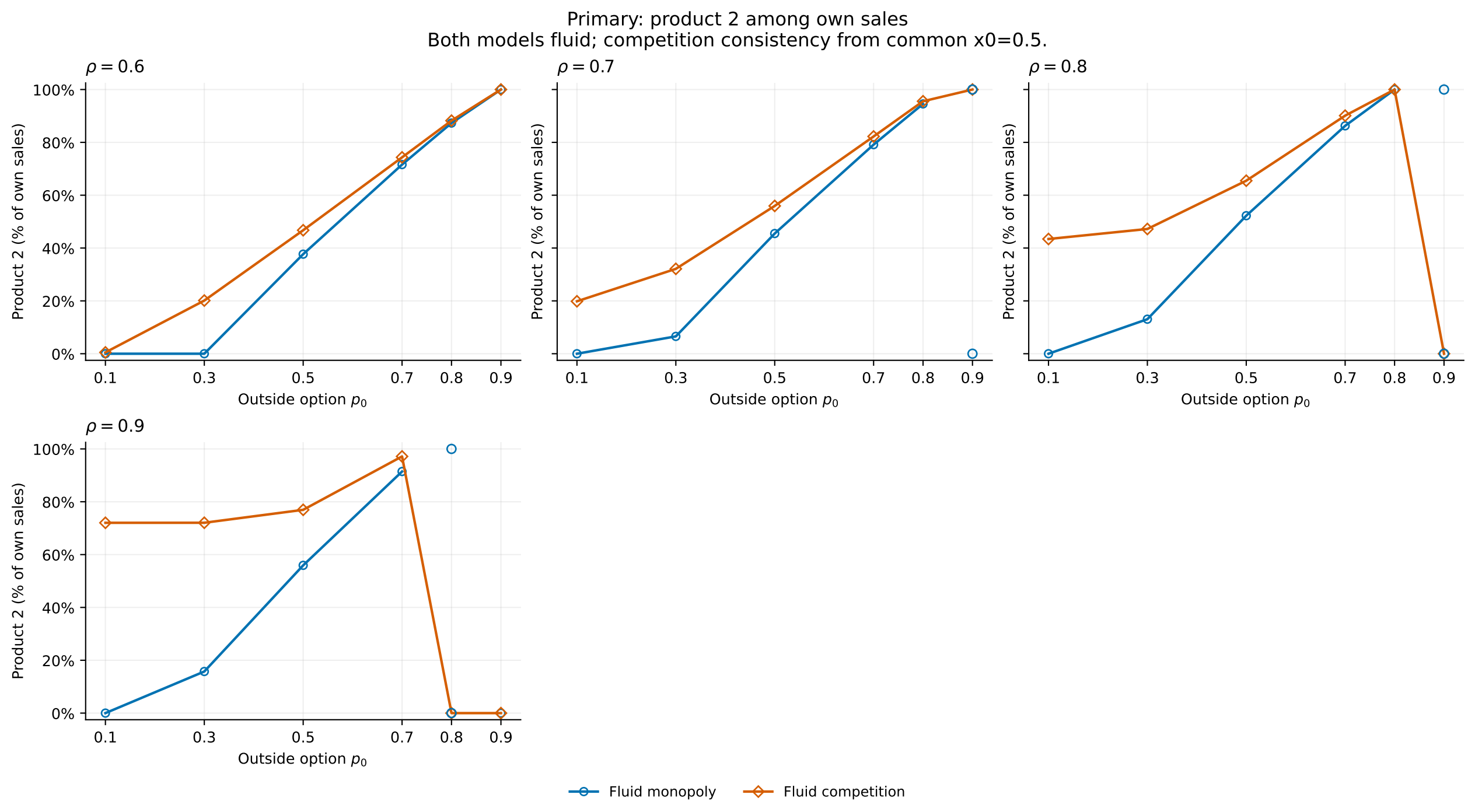
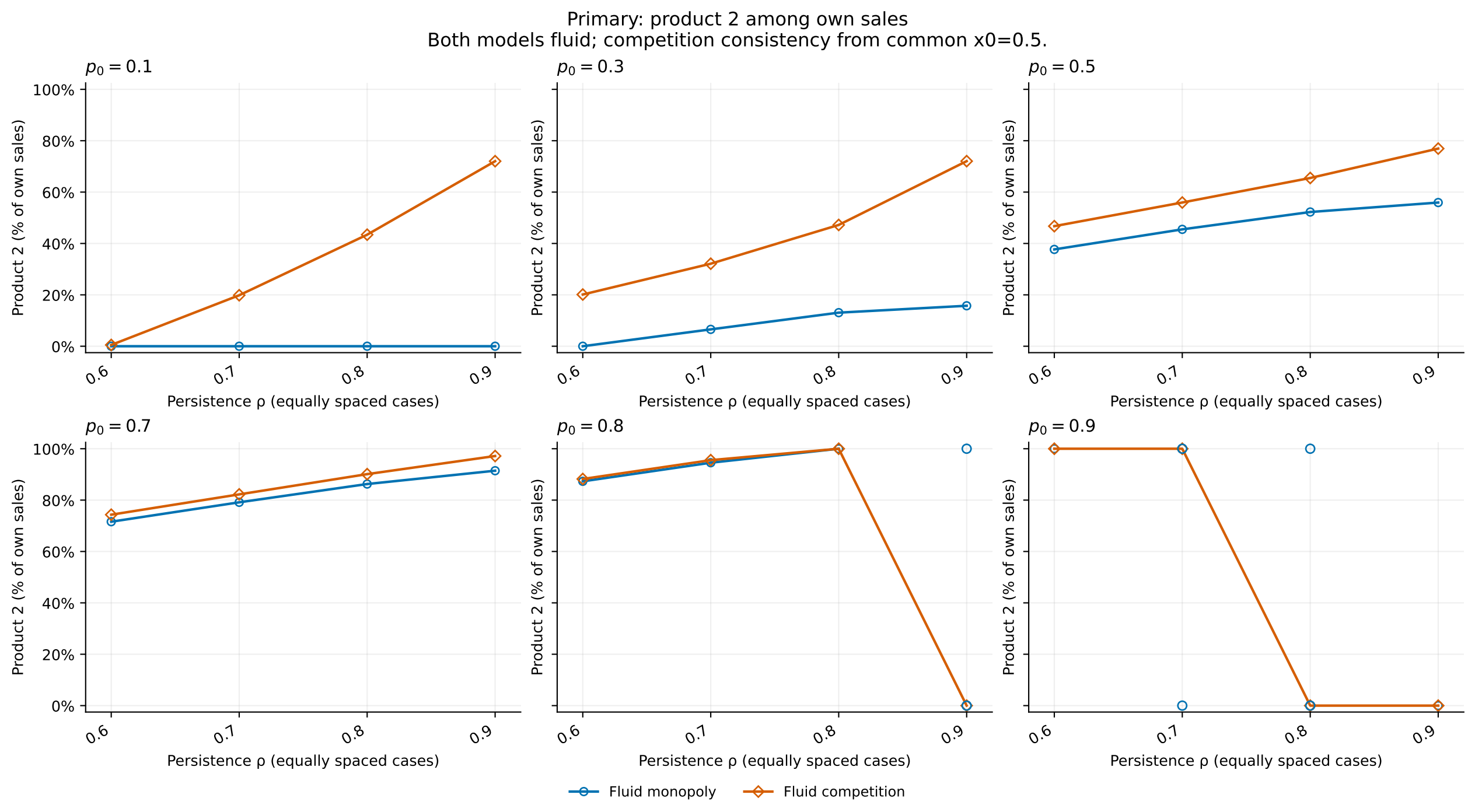
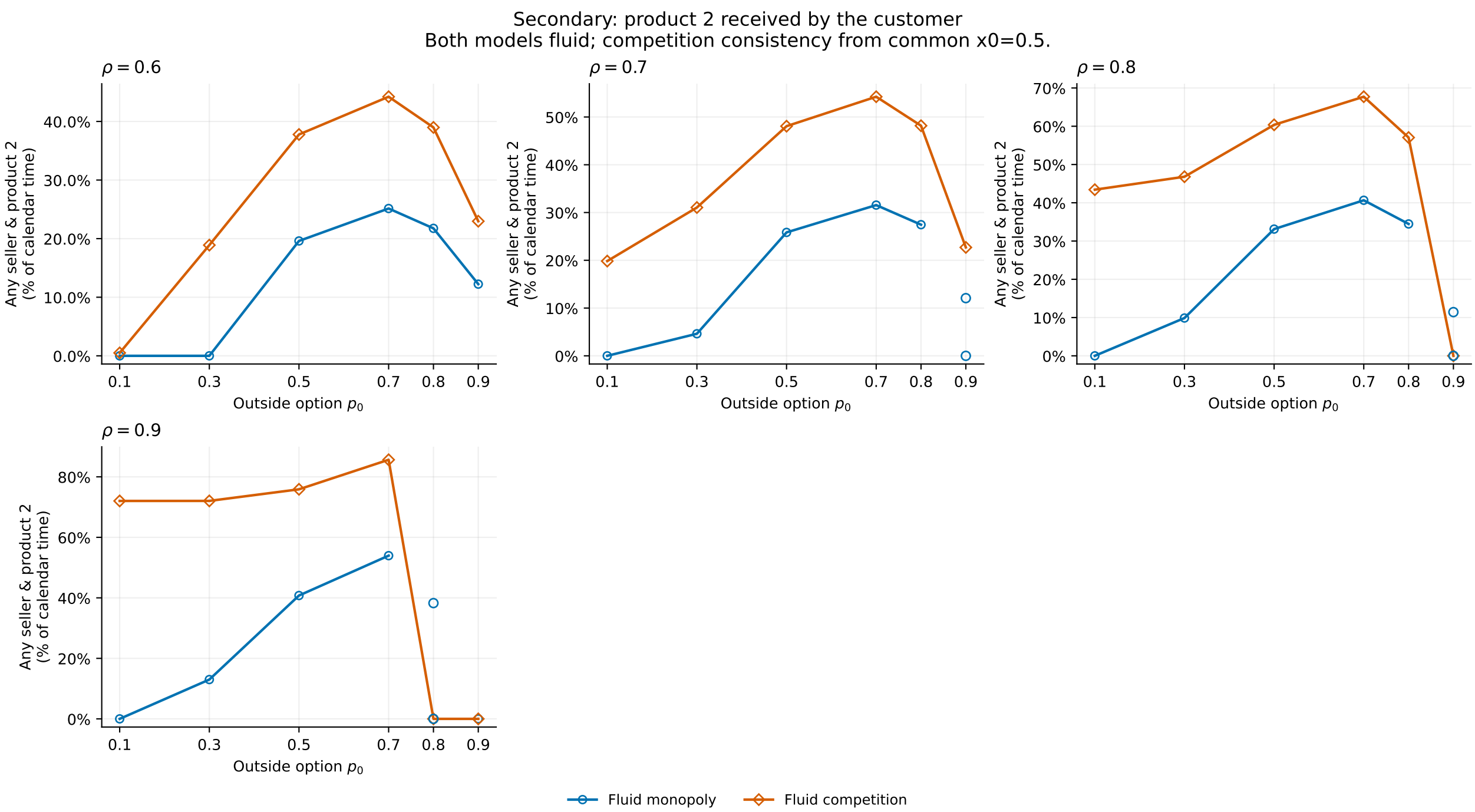
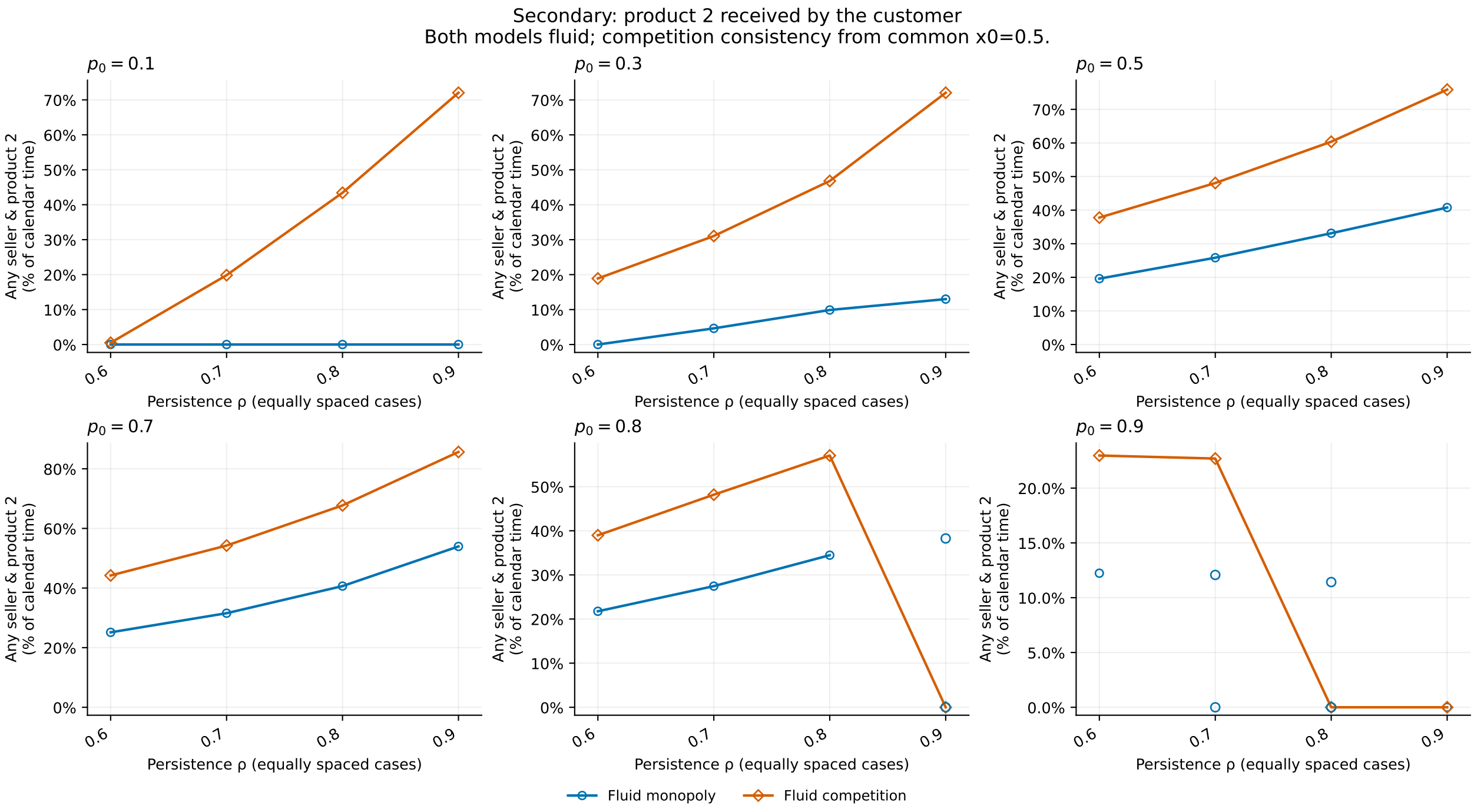
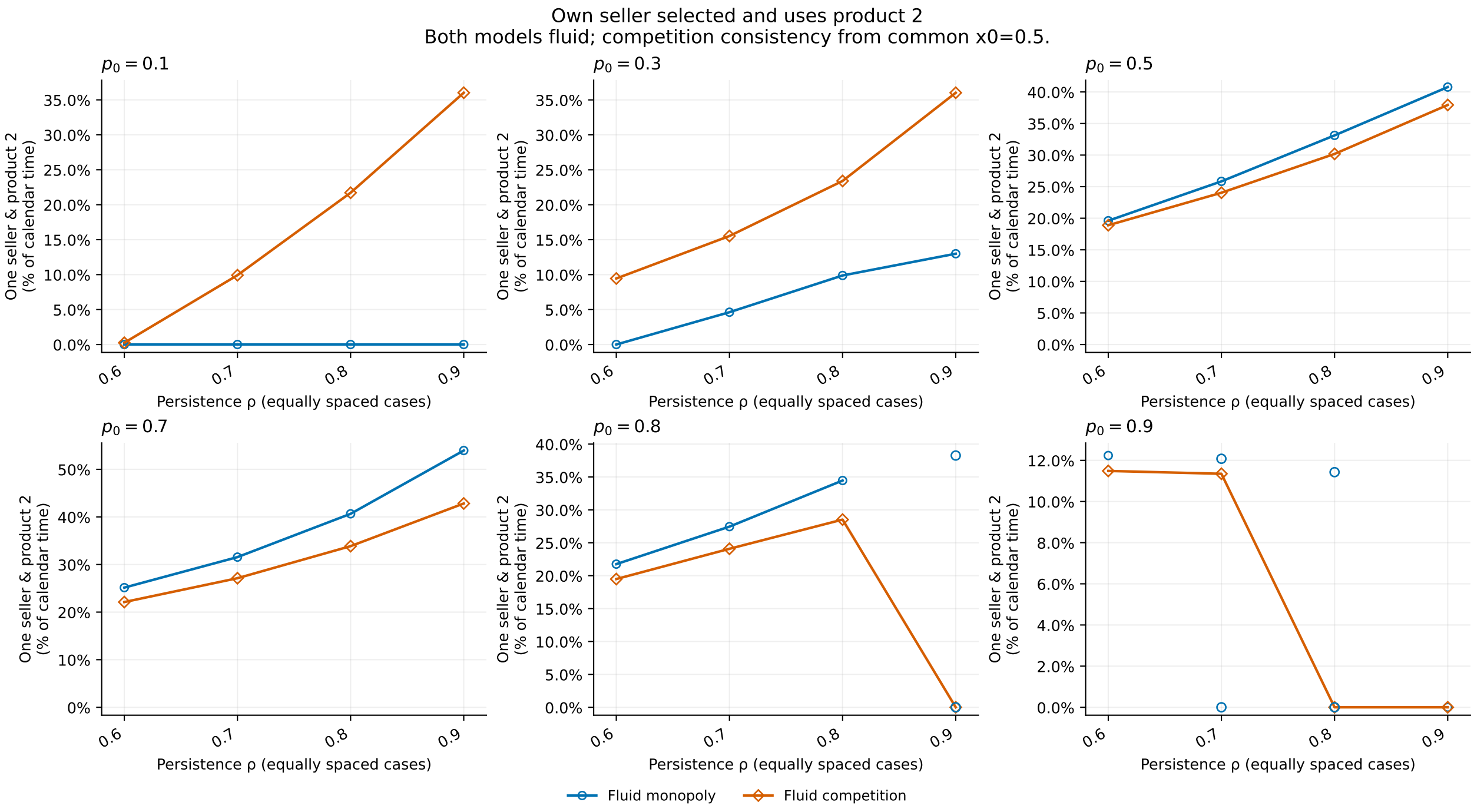
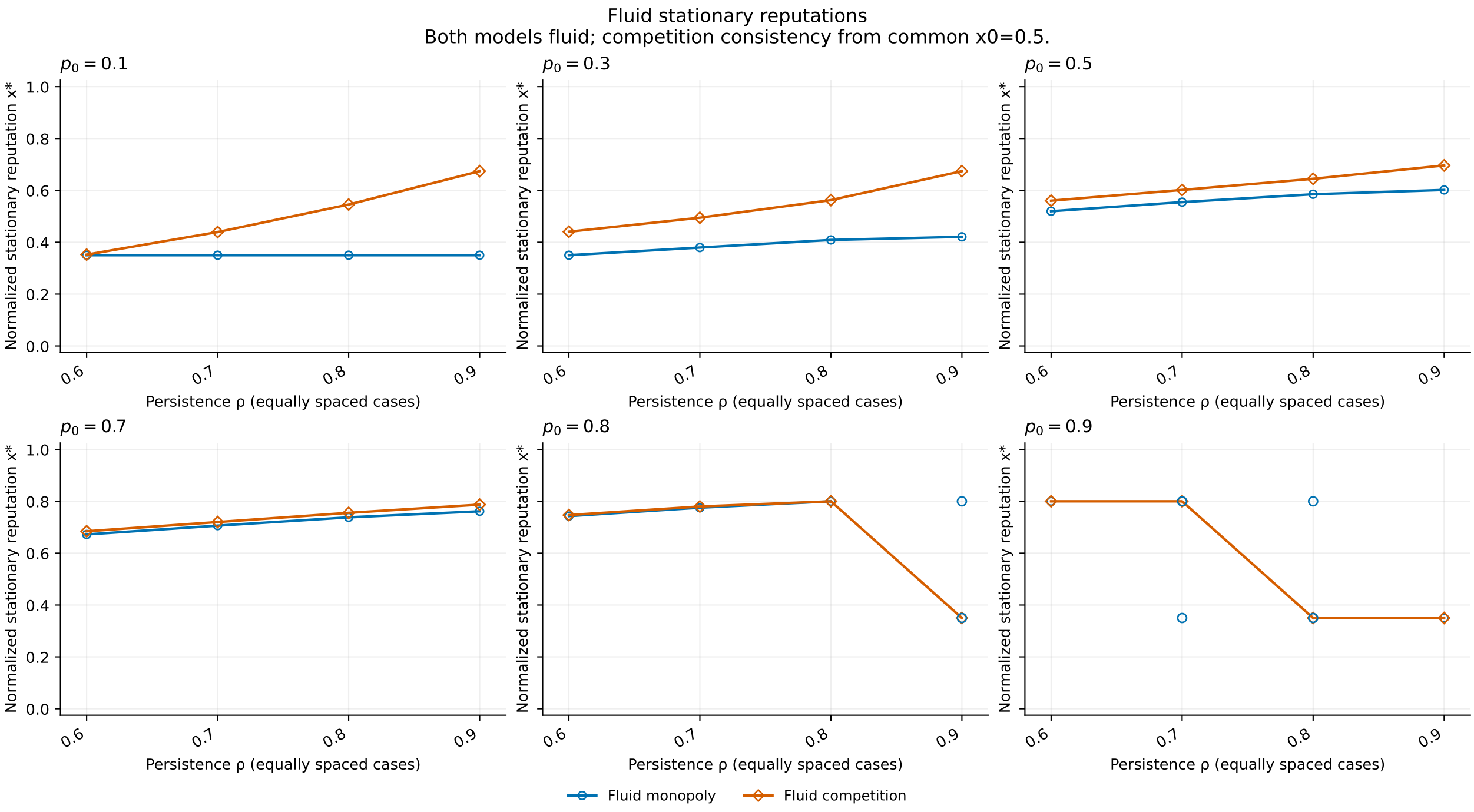

Primary: product 2 among own sales
Both models fluid; competition consistency from common x0=0.5.
ρ = 0.6
ρ = 0.7
ρ = 0.8
ρ = 0.9
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.1
0.3
0.5
0.7
0.8
0.9
Outside option p_0
Fluid monopoly
Fluid competition

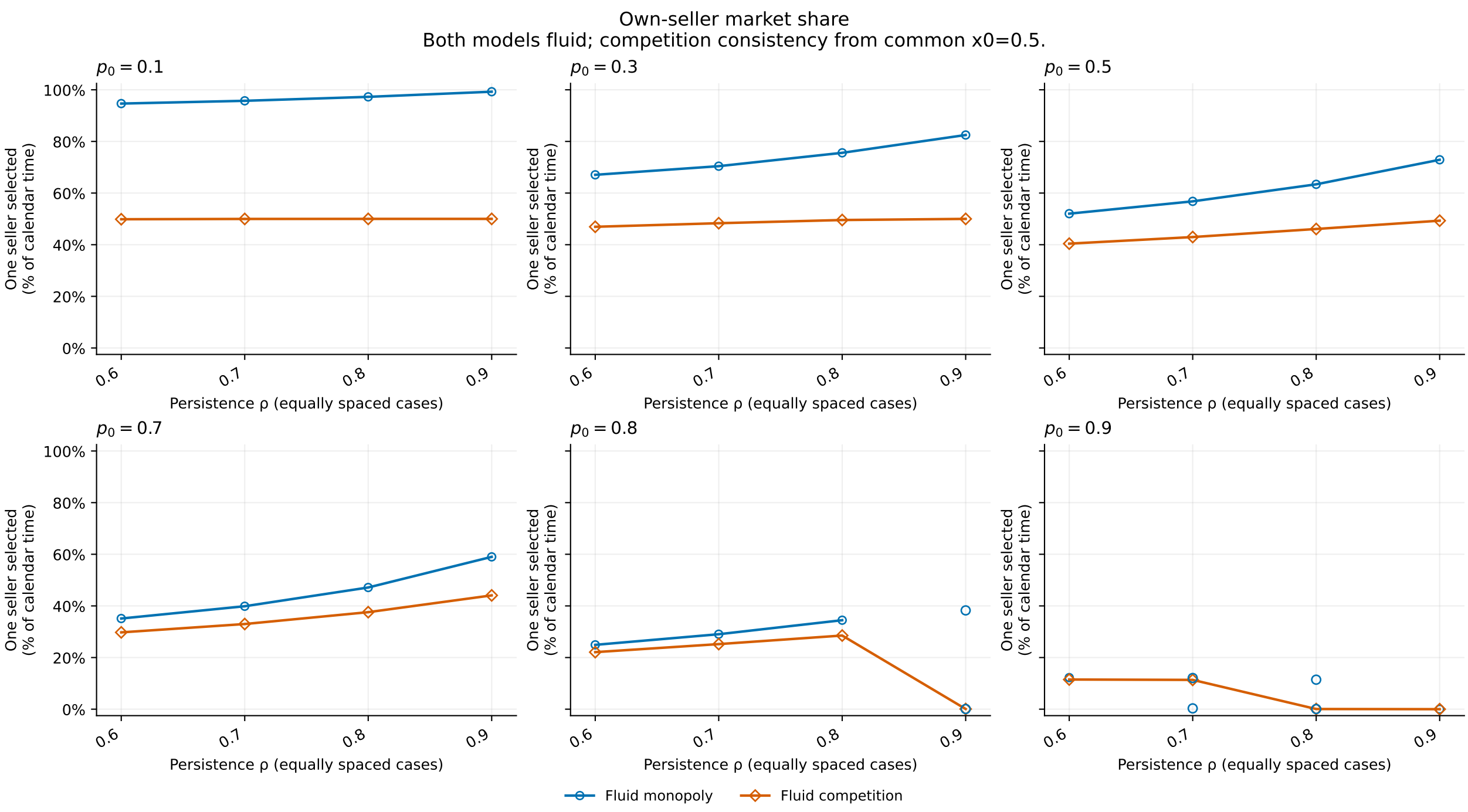

Primary: product 2 among own sales
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

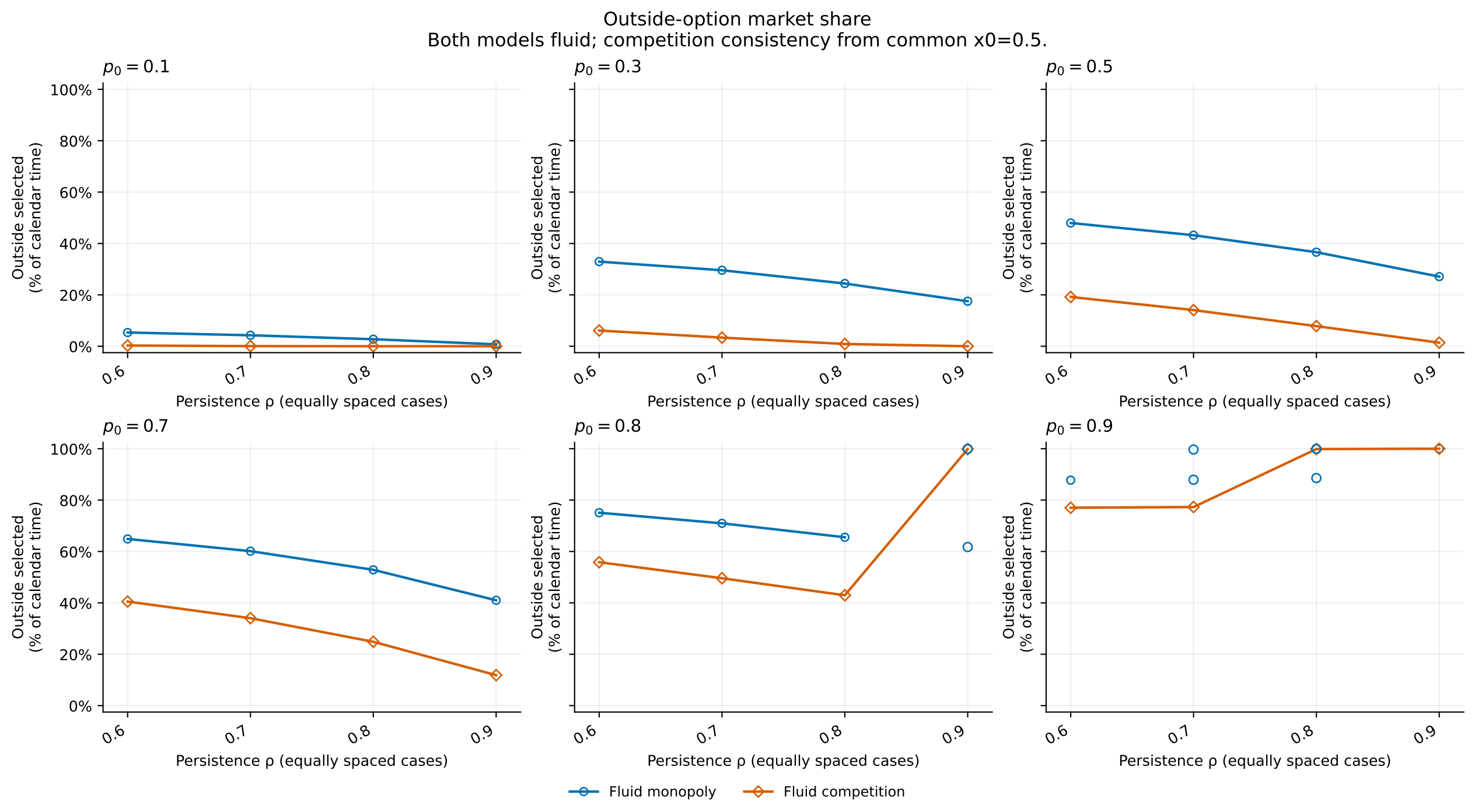

Secondary: product 2 received by the customer
Both models fluid; competition consistency from common x0=0.5.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
Any seller & product 2
(% of calendar time)
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
Fluid monopoly
Fluid competition

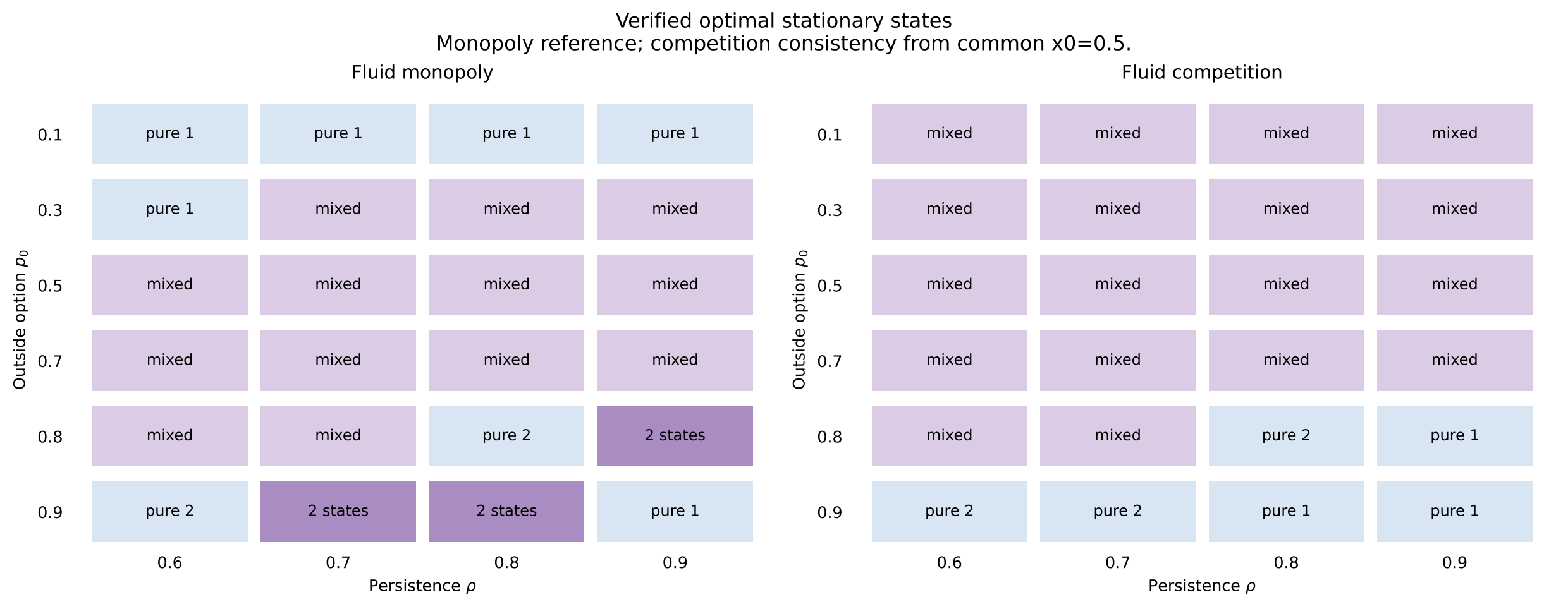

Secondary: product 2 received by the customer
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2
(% of calendar time)
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

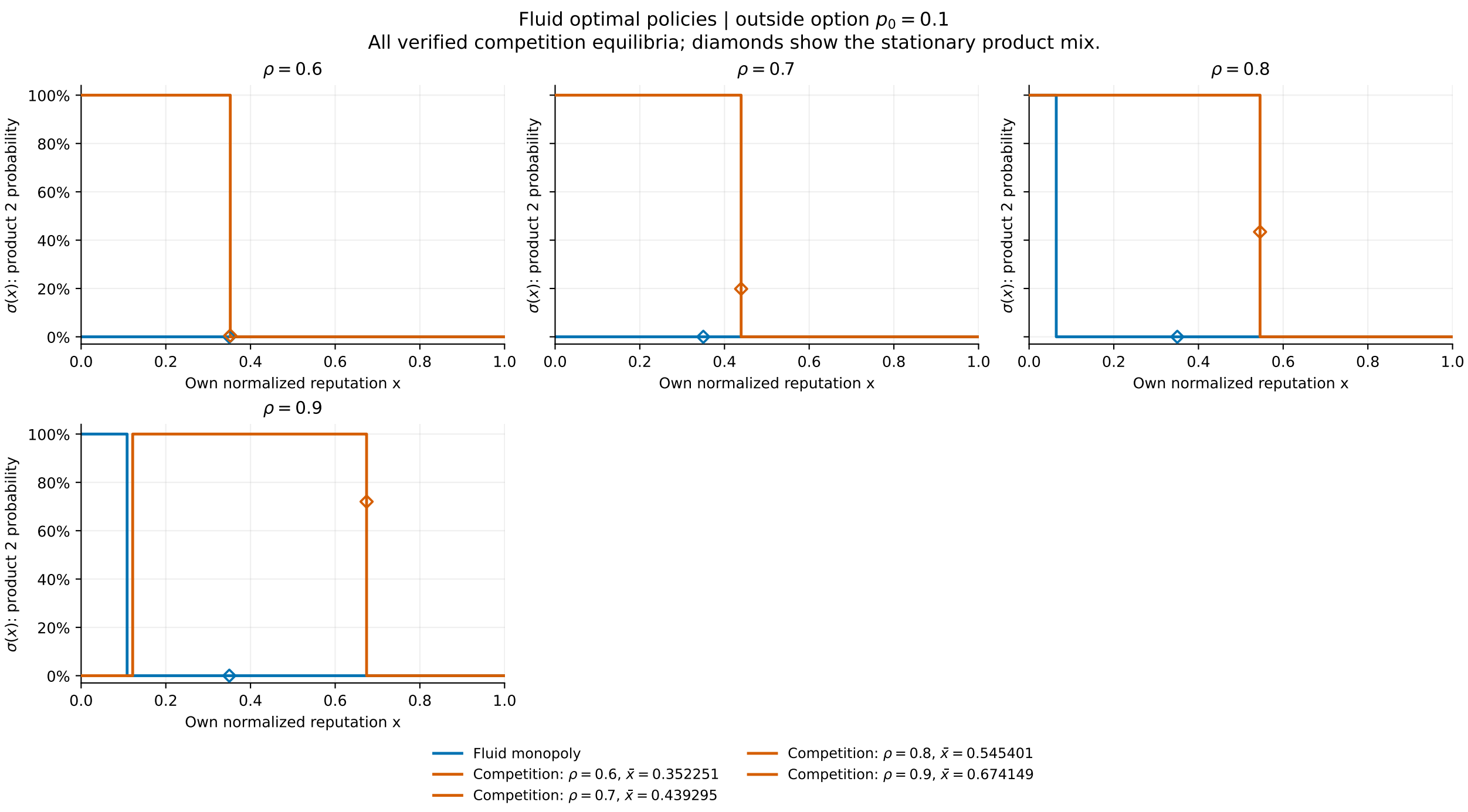

Own seller selected and uses product 2
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2
(% of calendar time)
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

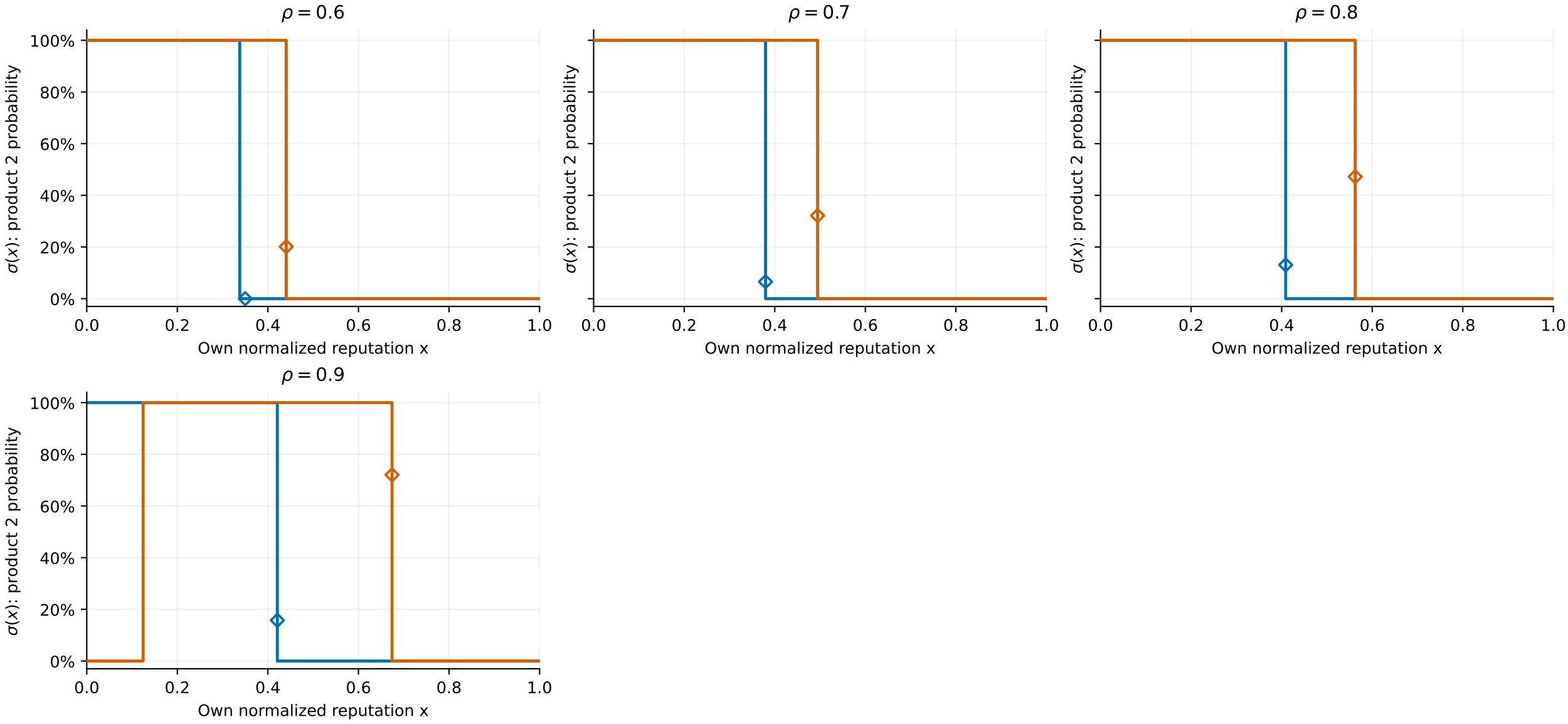

Fluid stationary reputations
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Normalized stationary reputation x*
Persistence ρ (equally spaced cases)
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.7
0.8
0.9
Fluid monopoly
Fluid competition

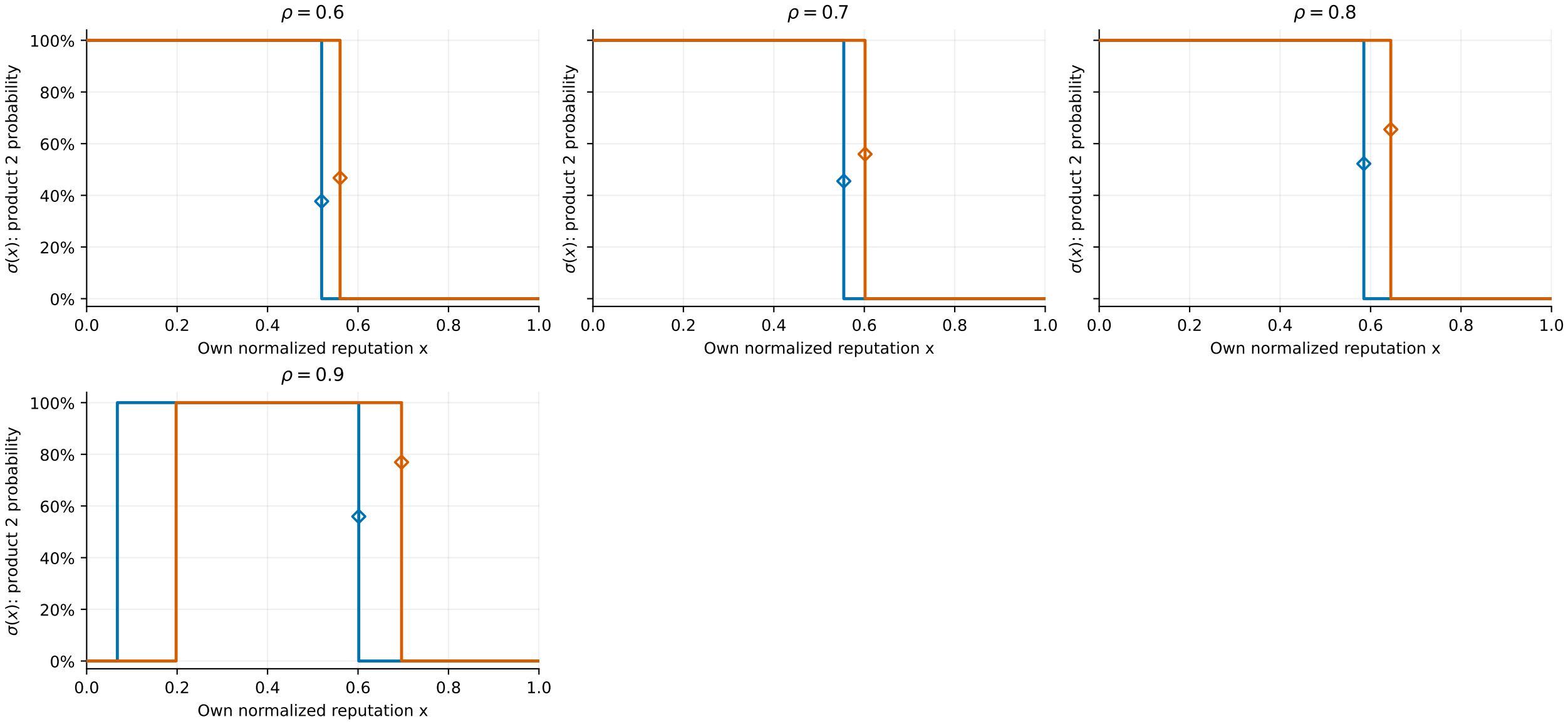

Own-seller market share
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller selected
(% of calendar time)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

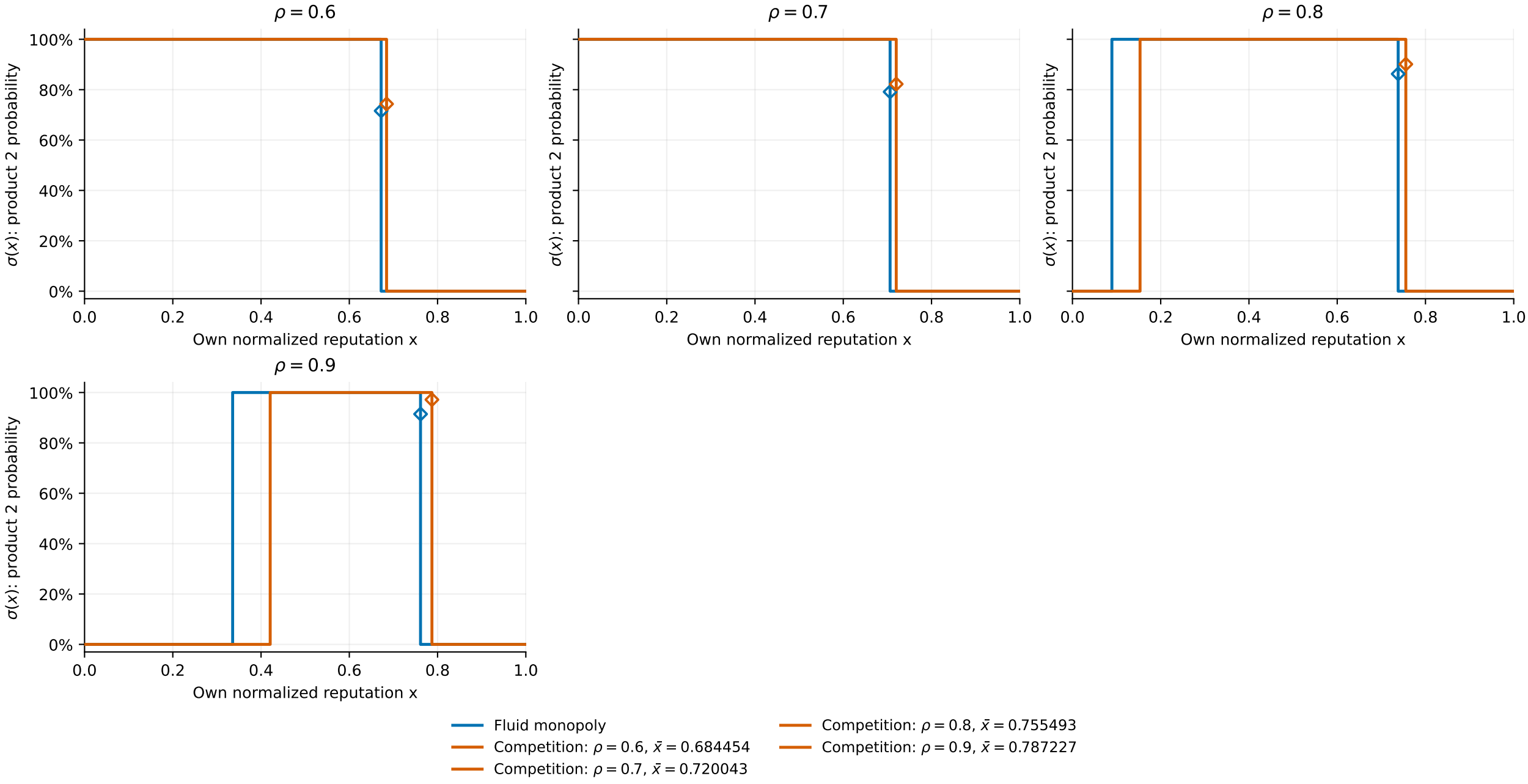

Outside-option market share
Both models fluid; competition consistency from common x0=0.5.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Outside selected (% of calendar time)
100%
80%
60%
40%
20%
0%
0.6
0.7
0.8
0.9
Persistence ρ (equally spaced cases)
Fluid monopoly
Fluid competition

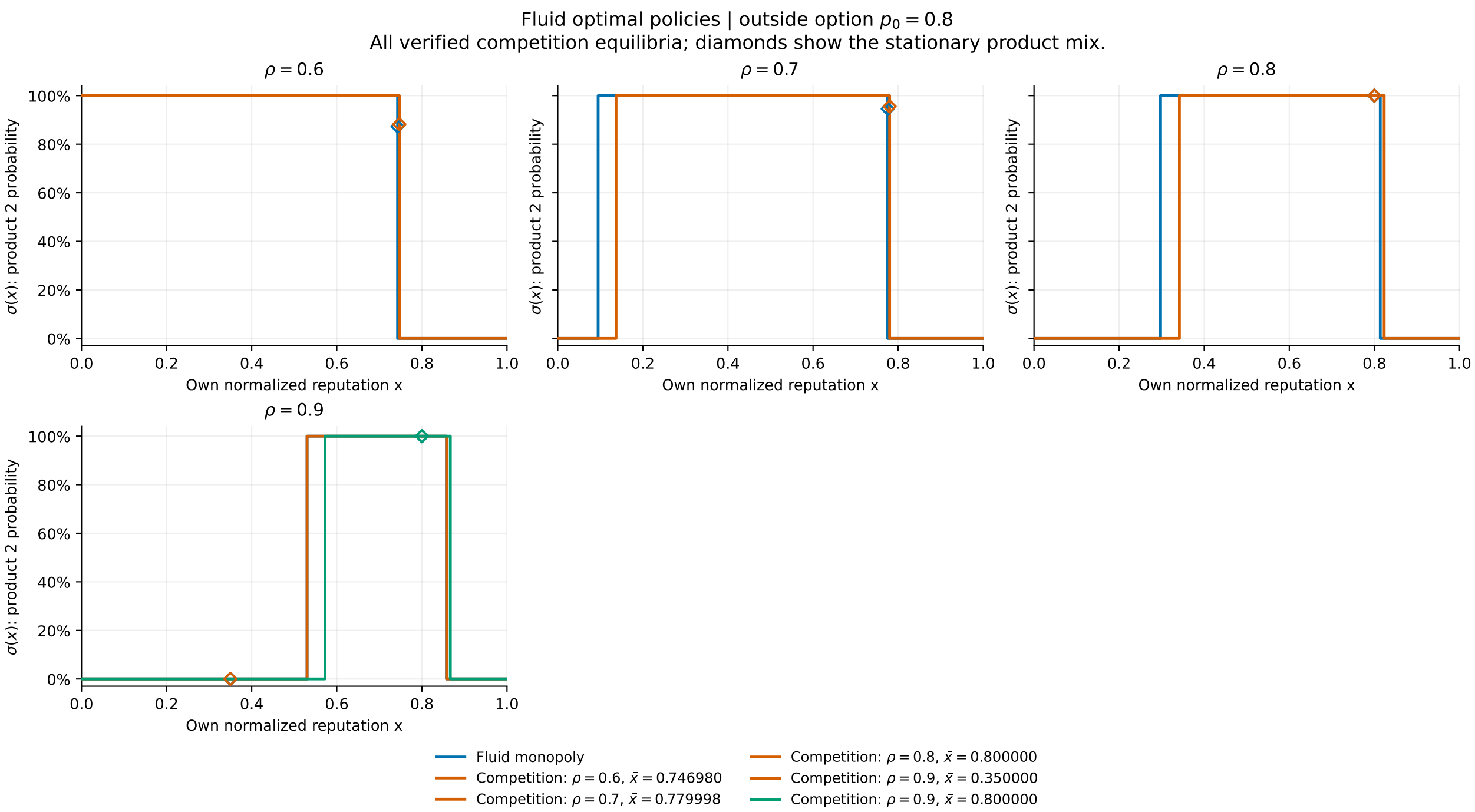

Verified optimal stationary states
Monopoly reference; competition consistency from common x0=0.5.
Fluid monopoly
Fluid competition
Outside option p_0
0.1
0.3
0.5
0.7
0.8
0.9
pure 1
pure 2
mixed
2 states
0.6
0.7
0.8
0.9
Persistence ρ

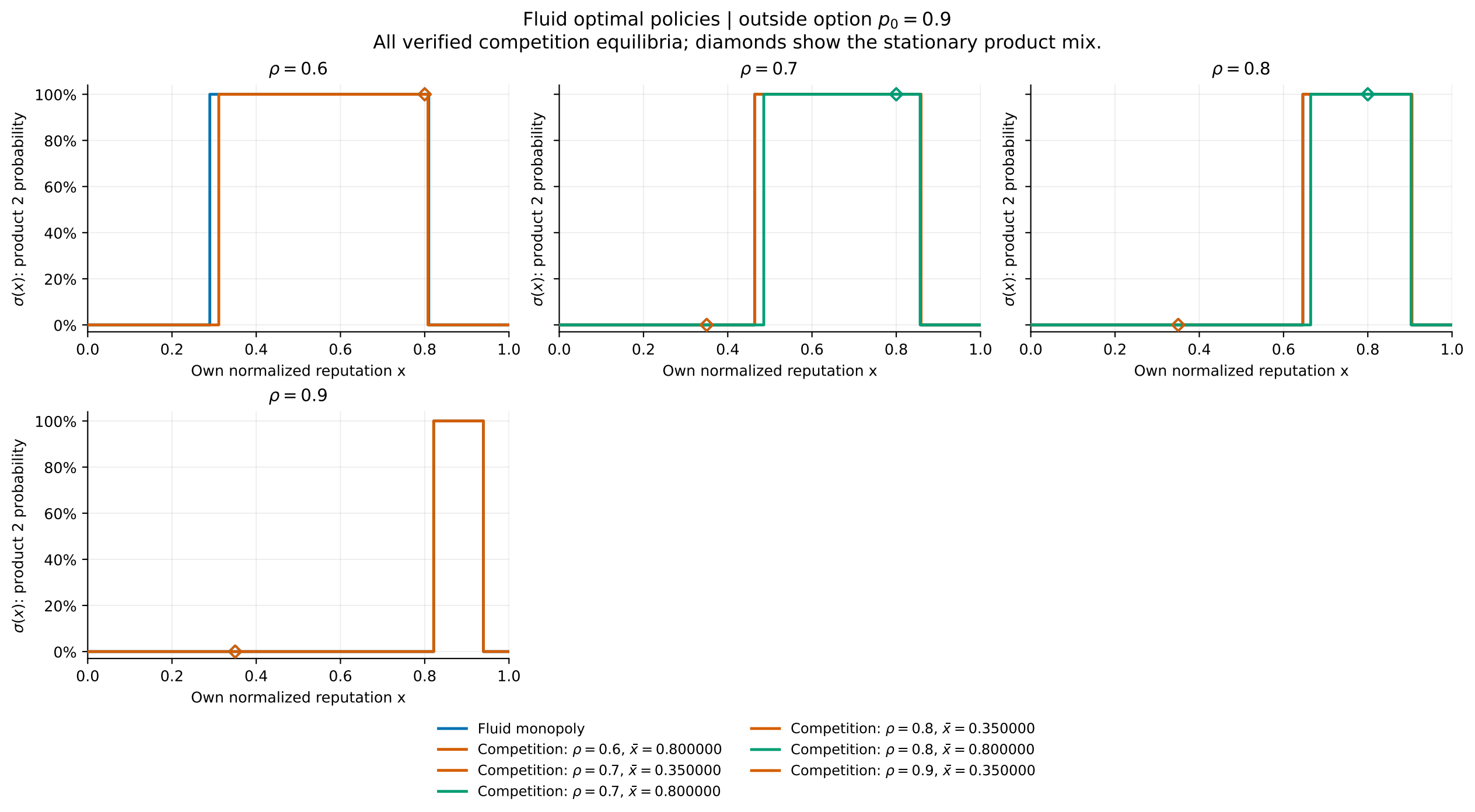

Fluid optimal policies | outside option $p_0 = 0.1$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.352251$
Competition: $\rho = 0.7$, $\bar{x} = 0.439295$
Competition: $\rho = 0.8$, $\bar{x} = 0.545401$
Competition: $\rho = 0.9$, $\bar{x} = 0.674149$

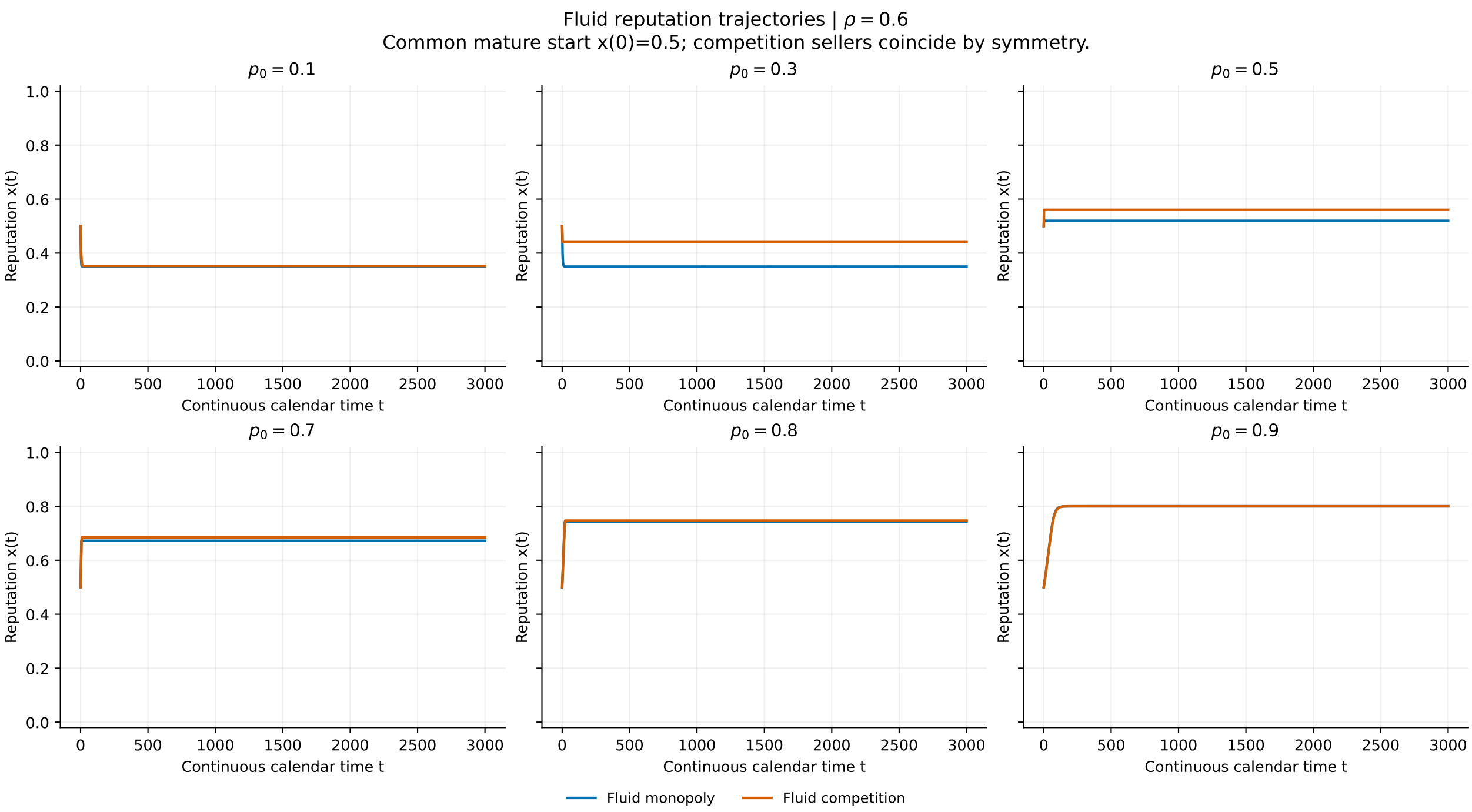

Fluid optimal policies | outside option $p_0 = 0.3$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.440565$
Competition: $\rho = 0.7$, $\bar{x} = 0.494567$
Competition: $\rho = 0.8$, $\bar{x} = 0.562469$
Competition: $\rho = 0.9$, $\bar{x} = 0.674219$

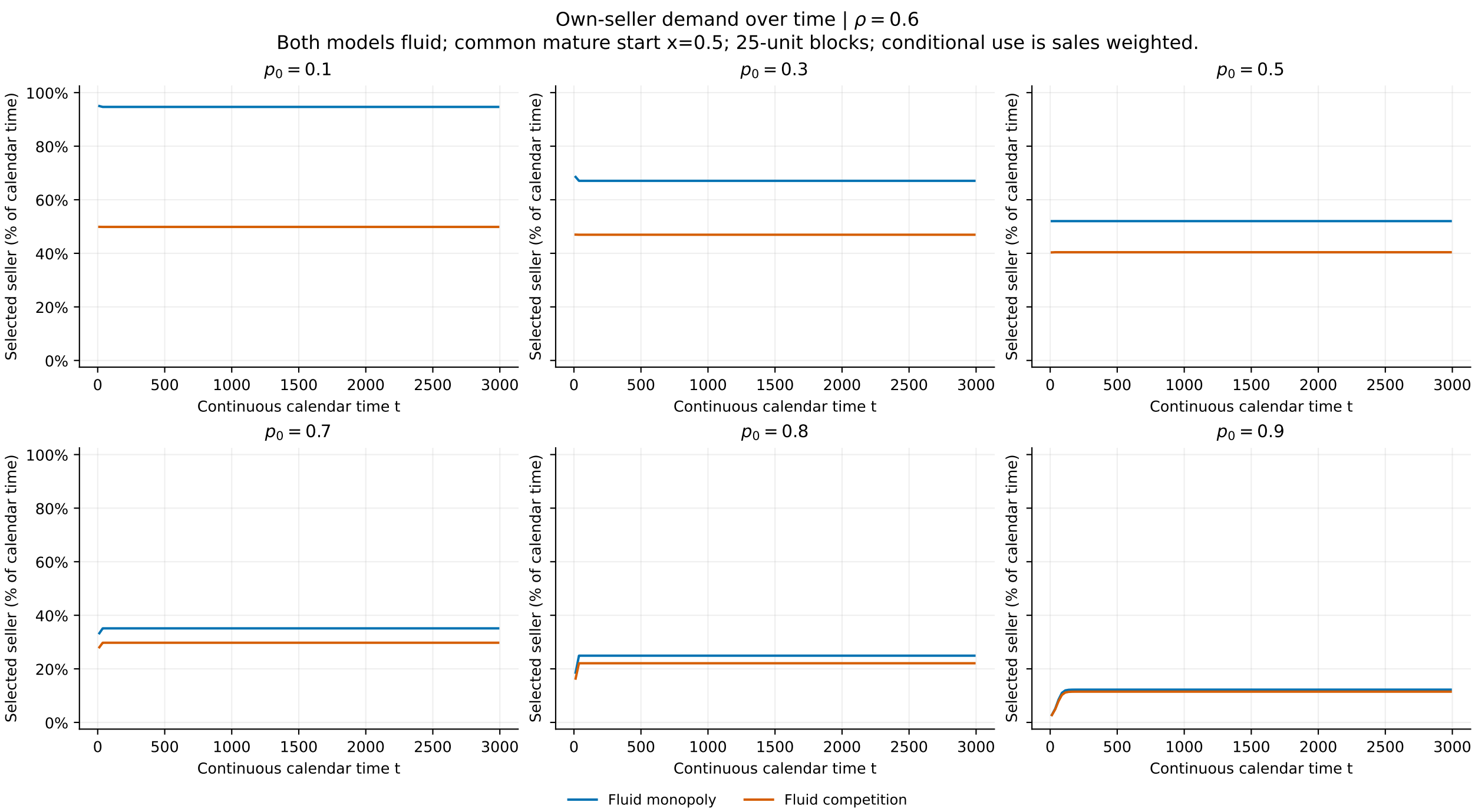

Fluid optimal policies | outside option $p_0 = 0.5$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.560326$
Competition: $\rho = 0.7$, $\bar{x} = 0.601750$
Competition: $\rho = 0.8$, $\bar{x} = 0.644722$
Competition: $\rho = 0.9$, $\bar{x} = 0.696244$

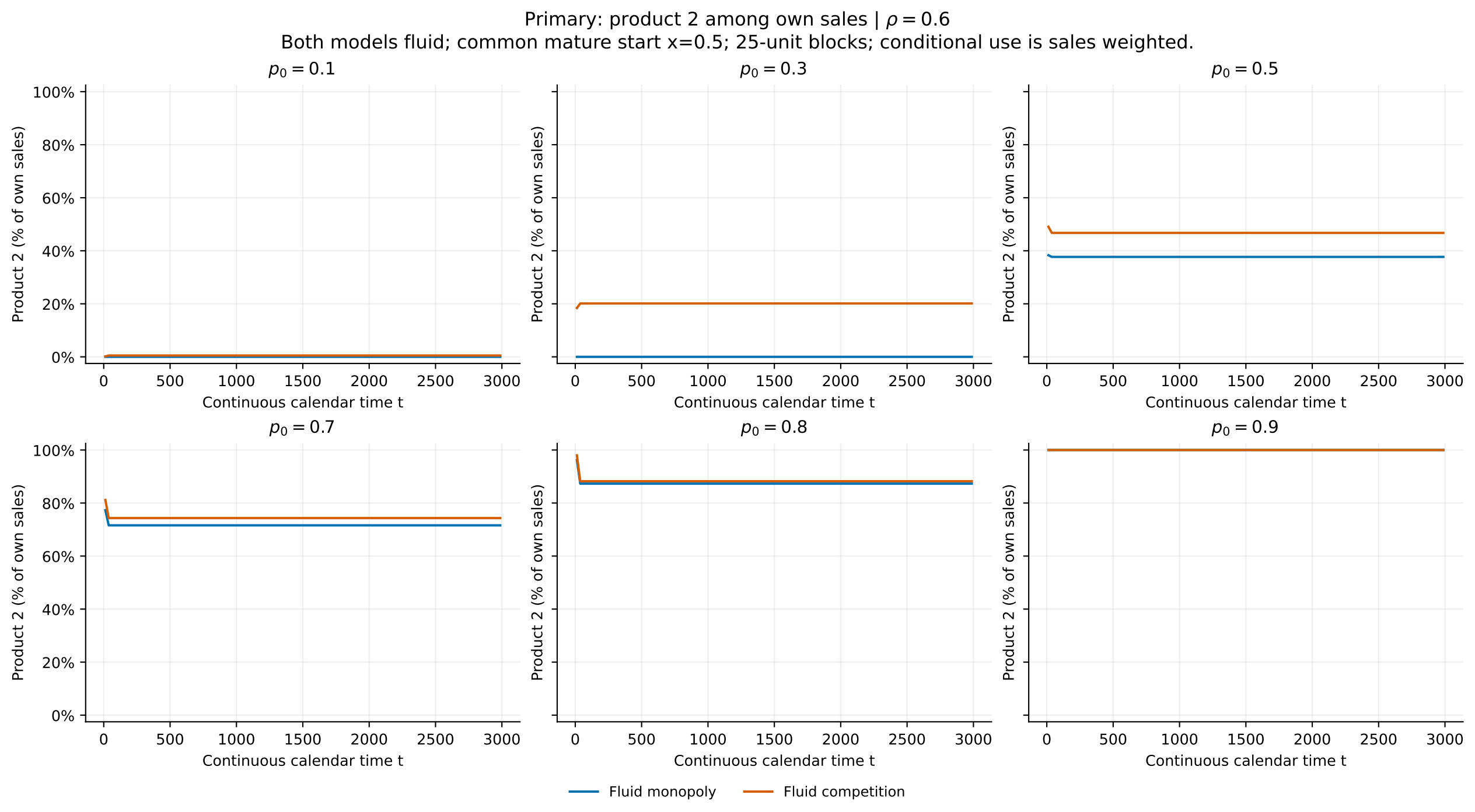

Fluid optimal policies | outside option $p_0 = 0.7$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.684454$
Competition: $\rho = 0.7$, $\bar{x} = 0.720043$
Competition: $\rho = 0.8$, $\bar{x} = 0.755493$
Competition: $\rho = 0.9$, $\bar{x} = 0.787227$

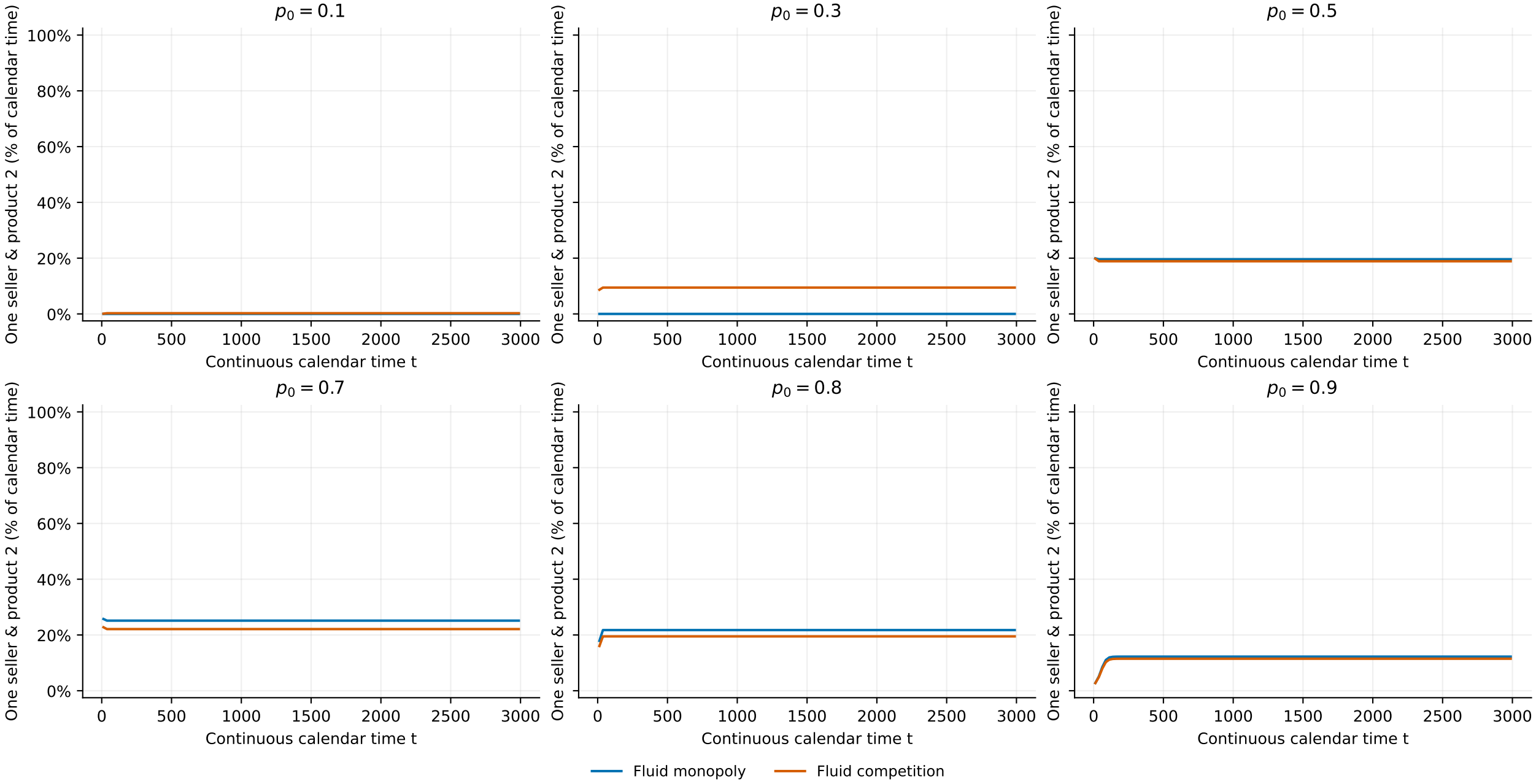

Fluid optimal policies | outside option $p_0 = 0.8$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.746980$
Competition: $\rho = 0.7$, $\bar{x} = 0.779998$
Competition: $\rho = 0.8$, $\bar{x} = 0.800000$
Competition: $\rho = 0.9$, $\bar{x} = 0.350000$
Competition: $\rho = 0.9$, $\bar{x} = 0.800000$

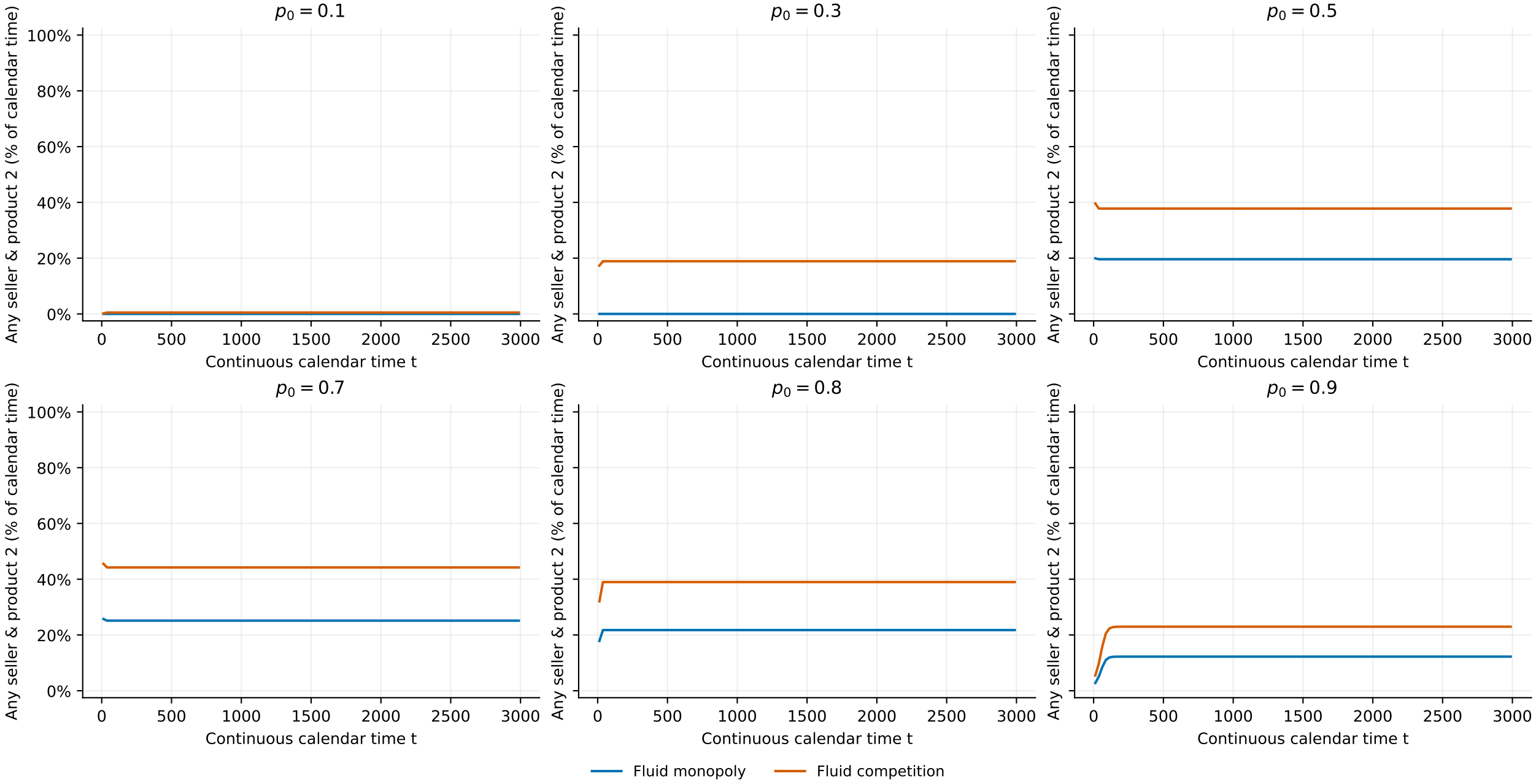

Fluid optimal policies | outside option $p_0 = 0.9$
All verified competition equilibria; diamonds show the stationary product mix.
$\rho = 0.6$
$\rho = 0.7$
$\rho = 0.8$
$\rho = 0.9$
$\sigma(x)$: product 2 probability
Own normalized reputation x
0%
20%
40%
60%
80%
100%
0.0
0.2
0.4
0.6
0.8
1.0
Fluid monopoly
Competition: $\rho = 0.6$, $\bar{x} = 0.800000$
Competition: $\rho = 0.7$, $\bar{x} = 0.350000$
Competition: $\rho = 0.7$, $\bar{x} = 0.800000$
Competition: $\rho = 0.8$, $\bar{x} = 0.350000$
Competition: $\rho = 0.8$, $\bar{x} = 0.800000$
Competition: $\rho = 0.9$, $\bar{x} = 0.350000$

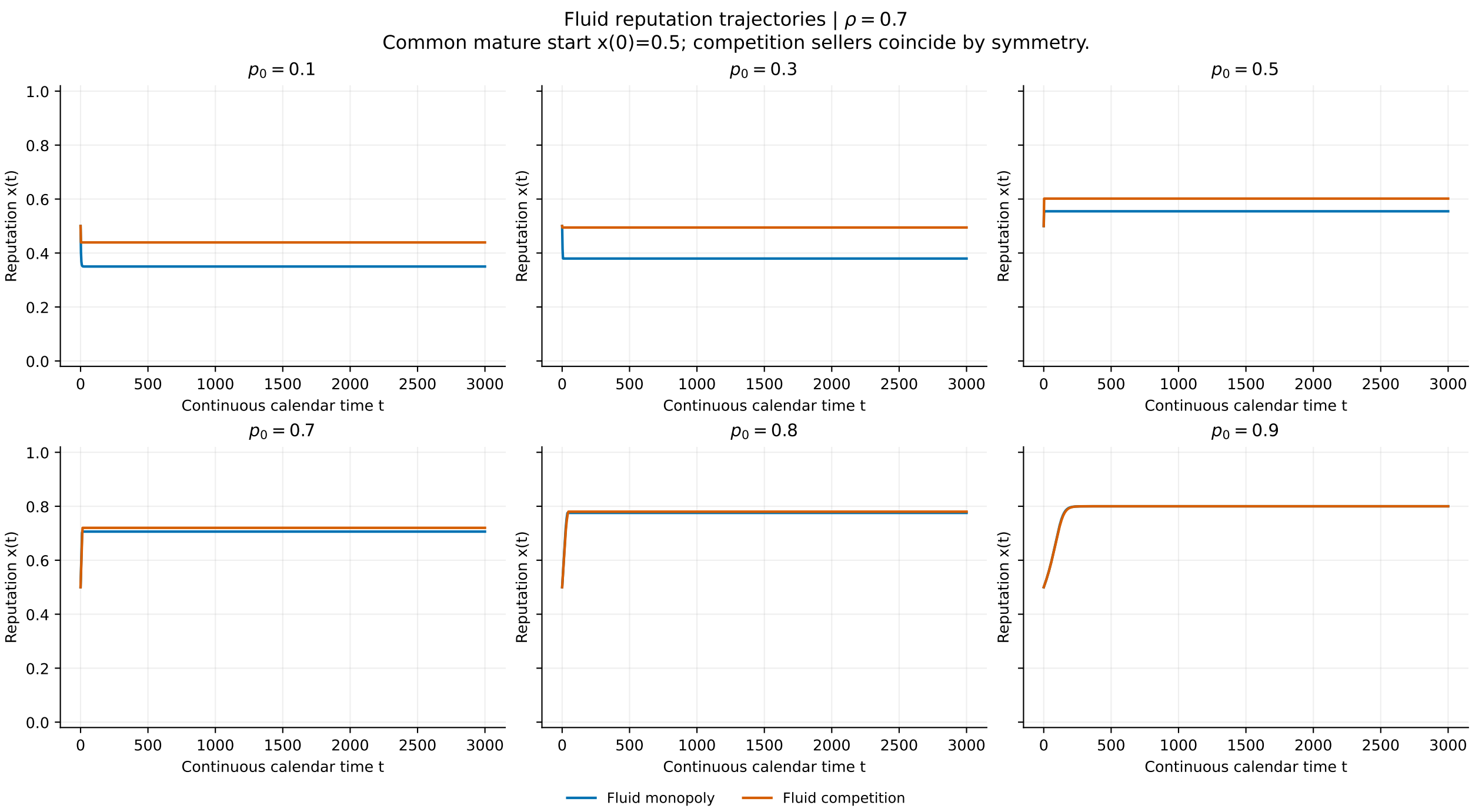

Fluid reputation trajectories | $\rho = 0.6$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

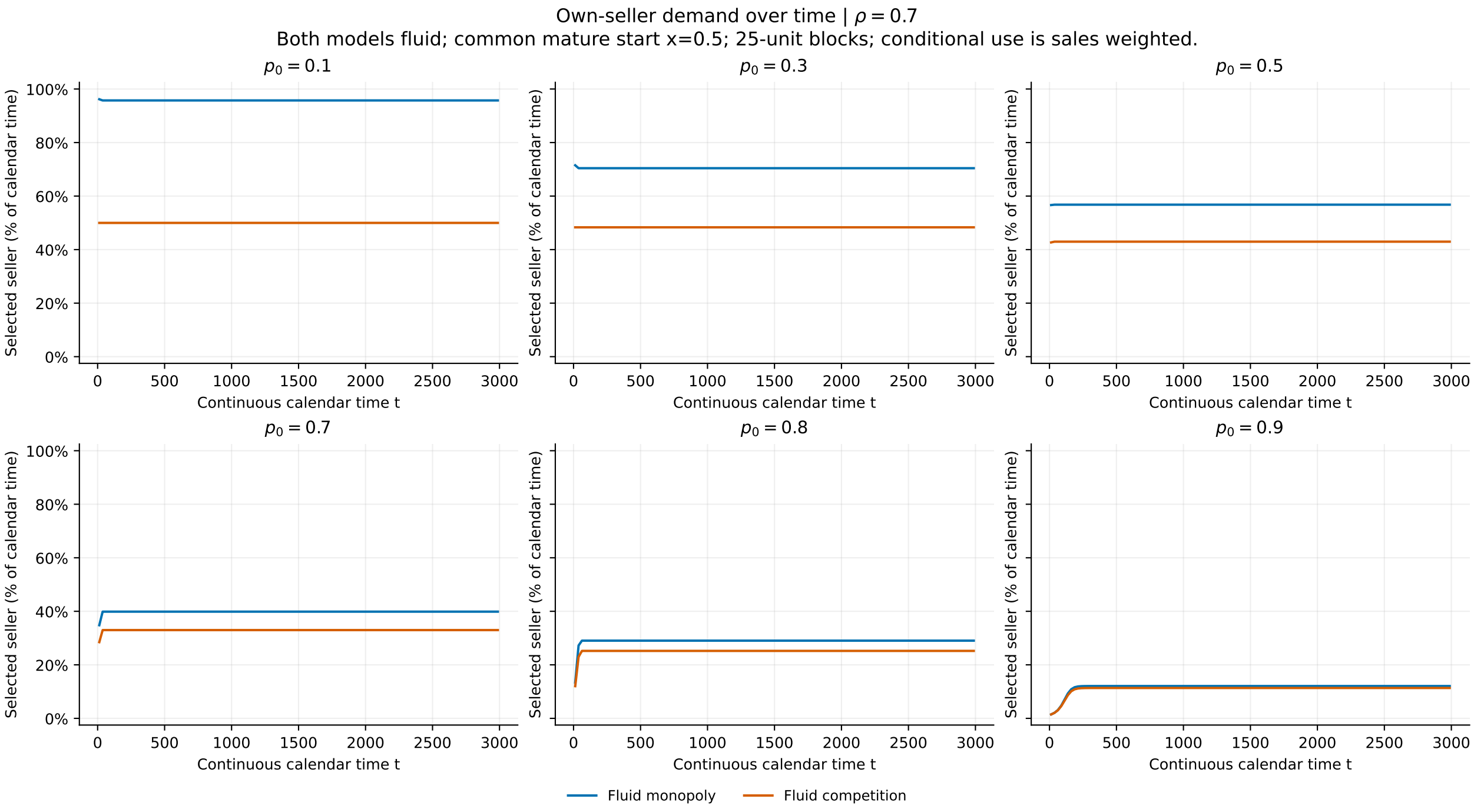

Own-seller demand over time | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

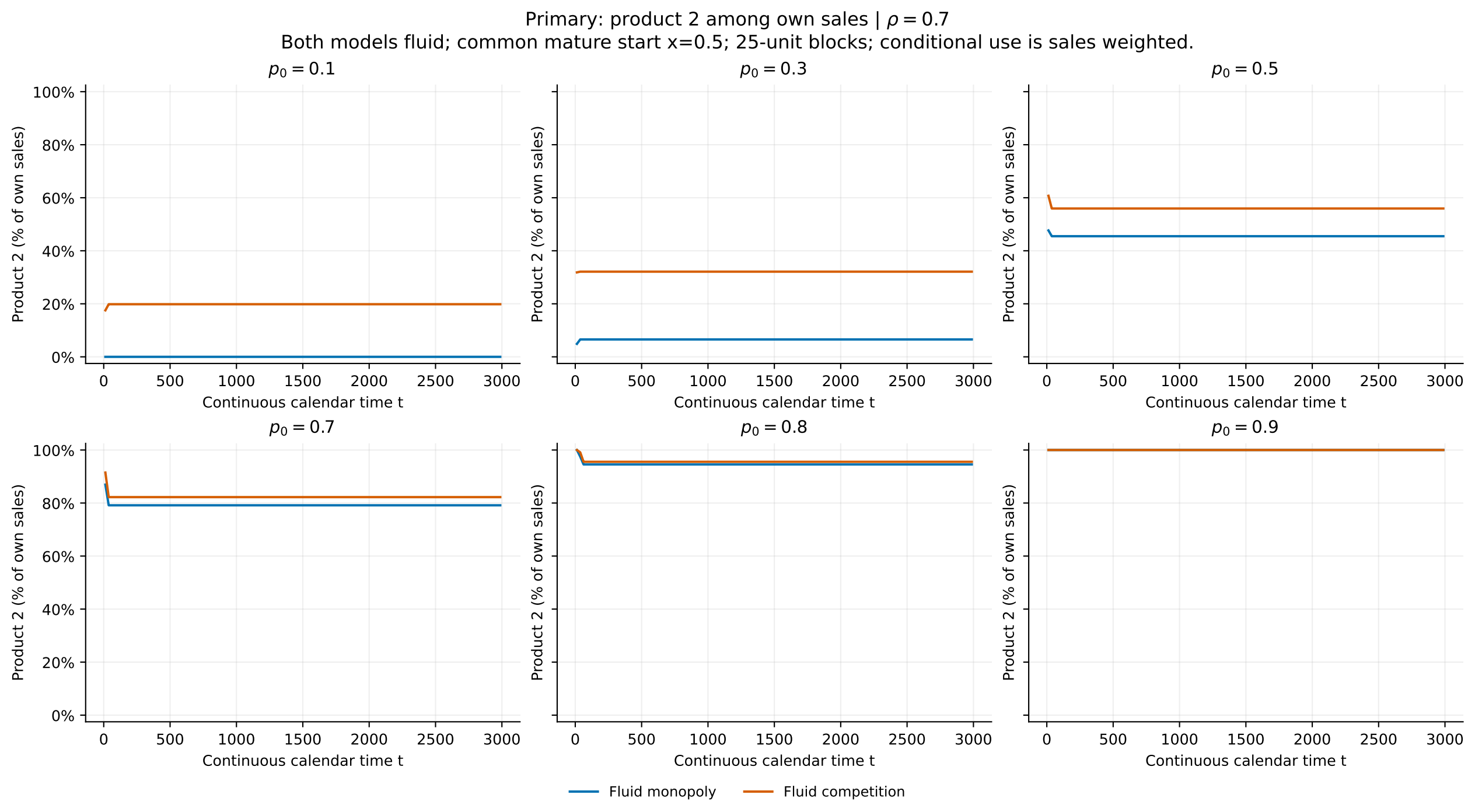

Primary: product 2 among own sales | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

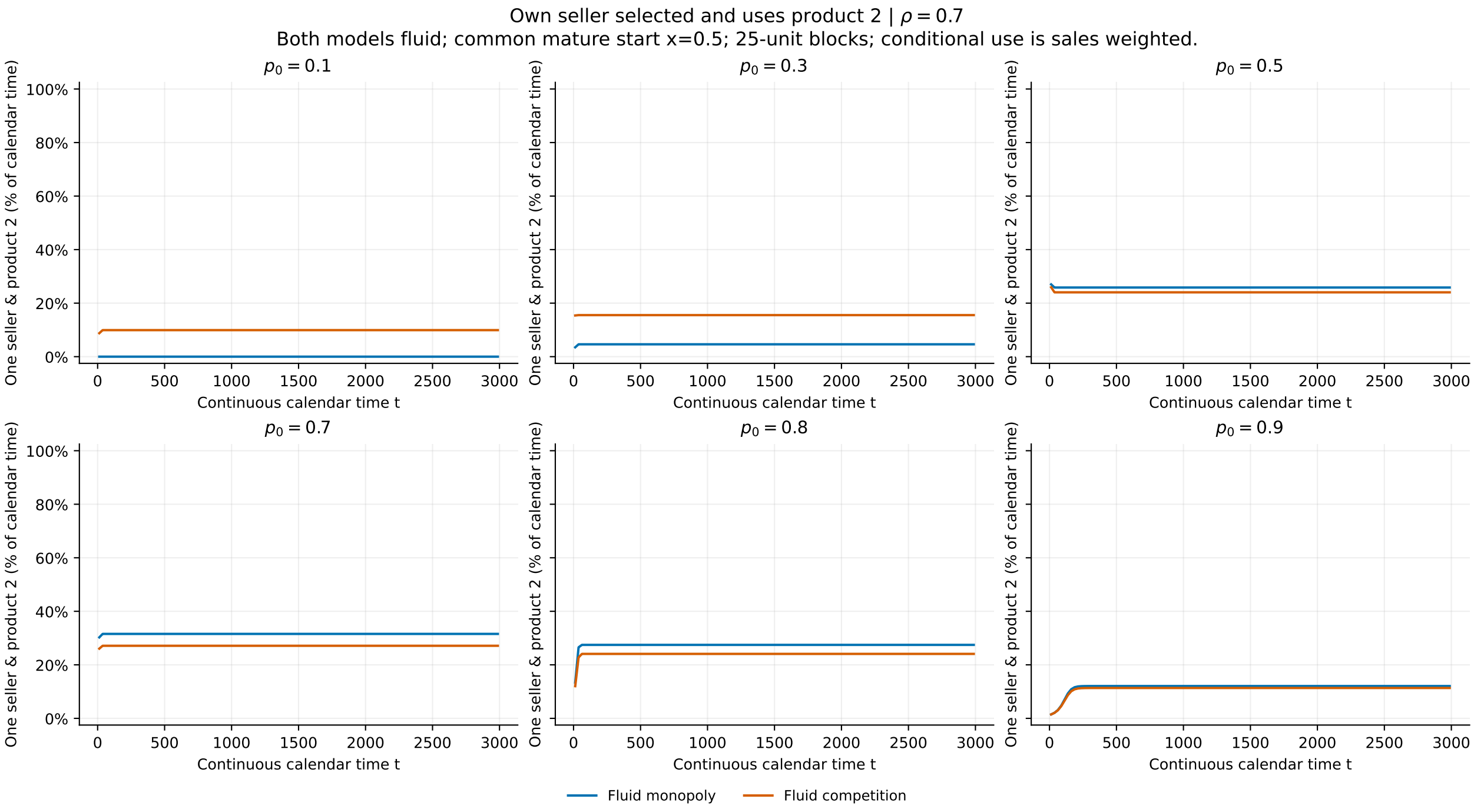

Own seller selected and uses product 2 | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

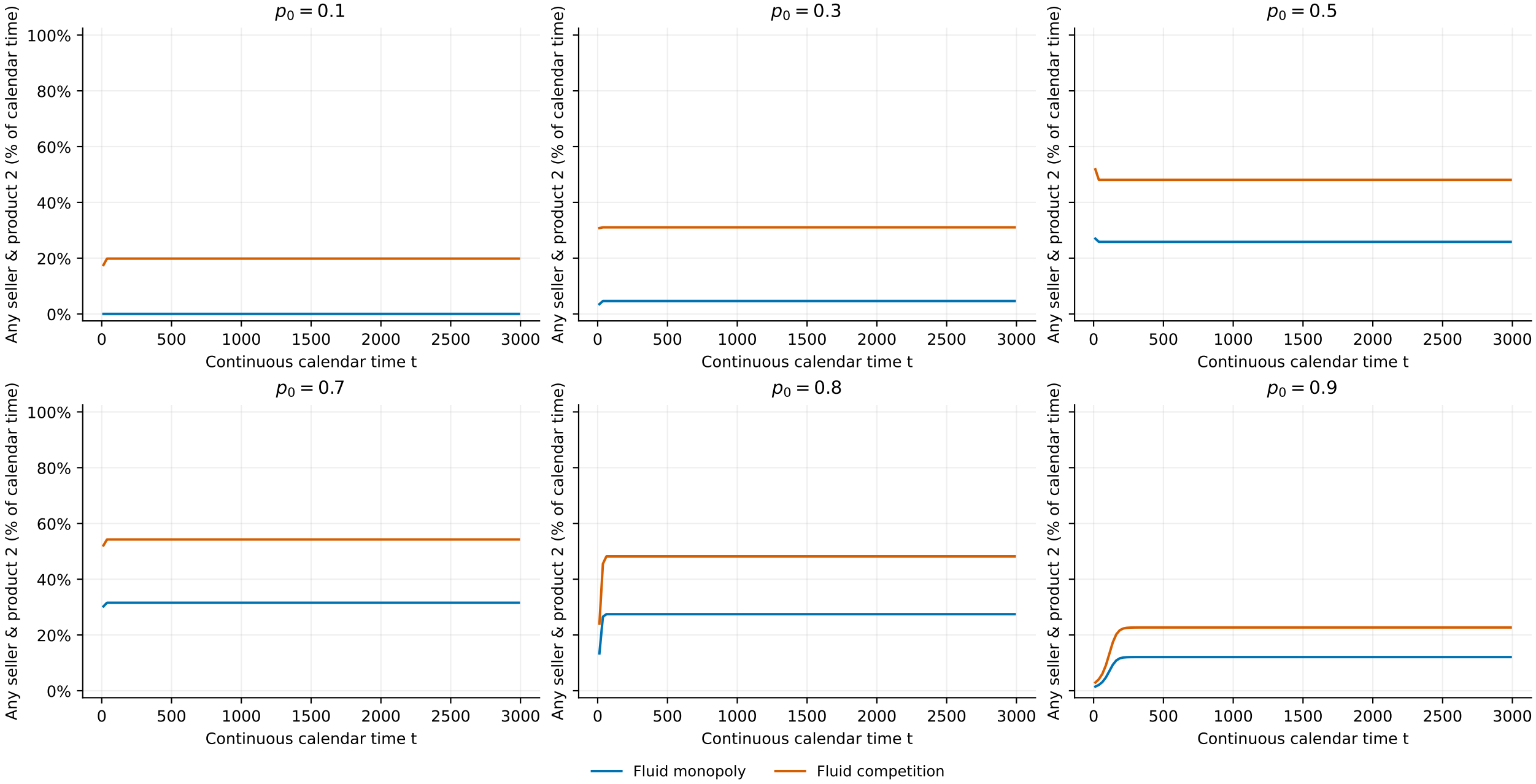

Secondary: customer receives product 2 | $\rho = 0.6$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

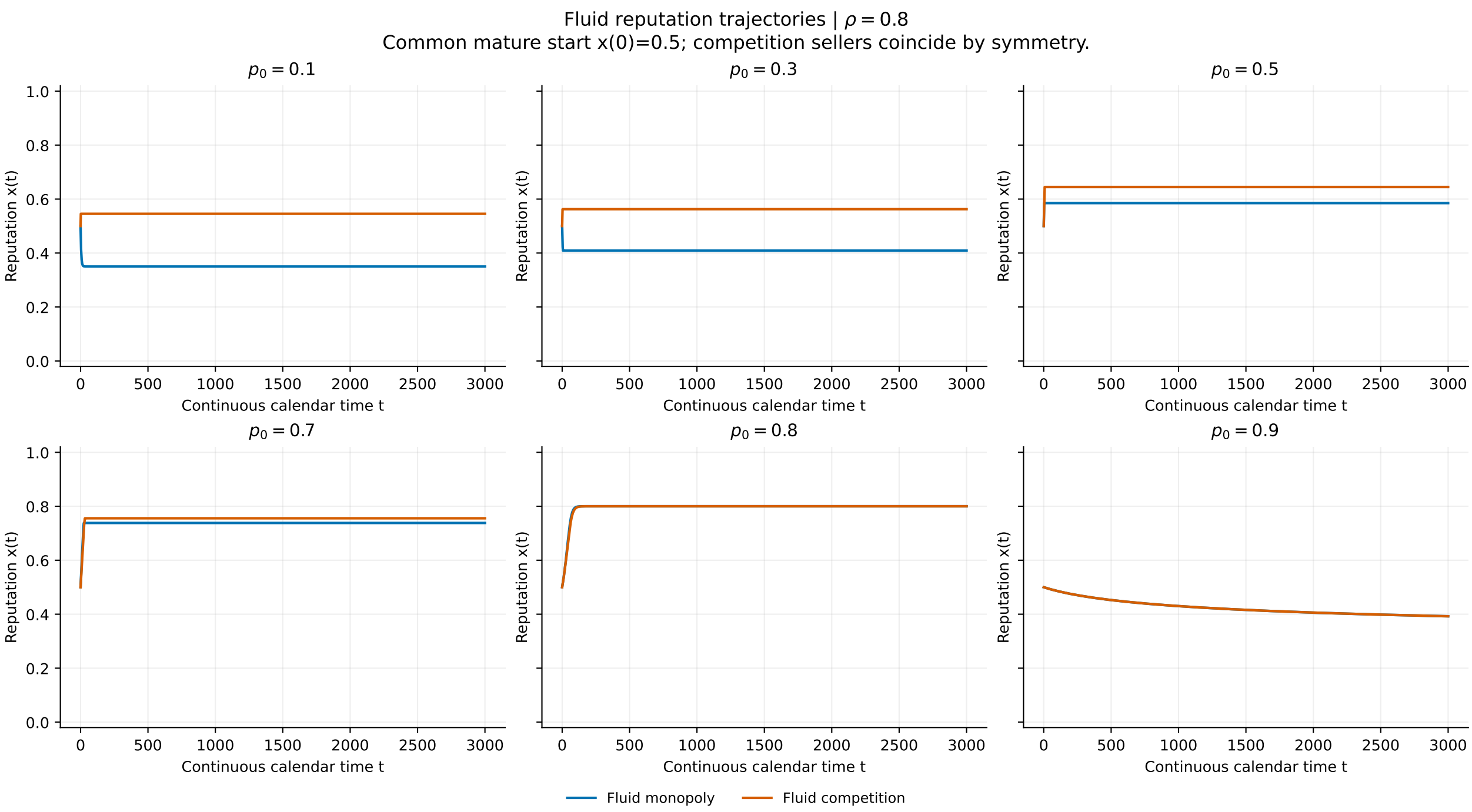

Fluid reputation trajectories | $\rho = 0.7$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

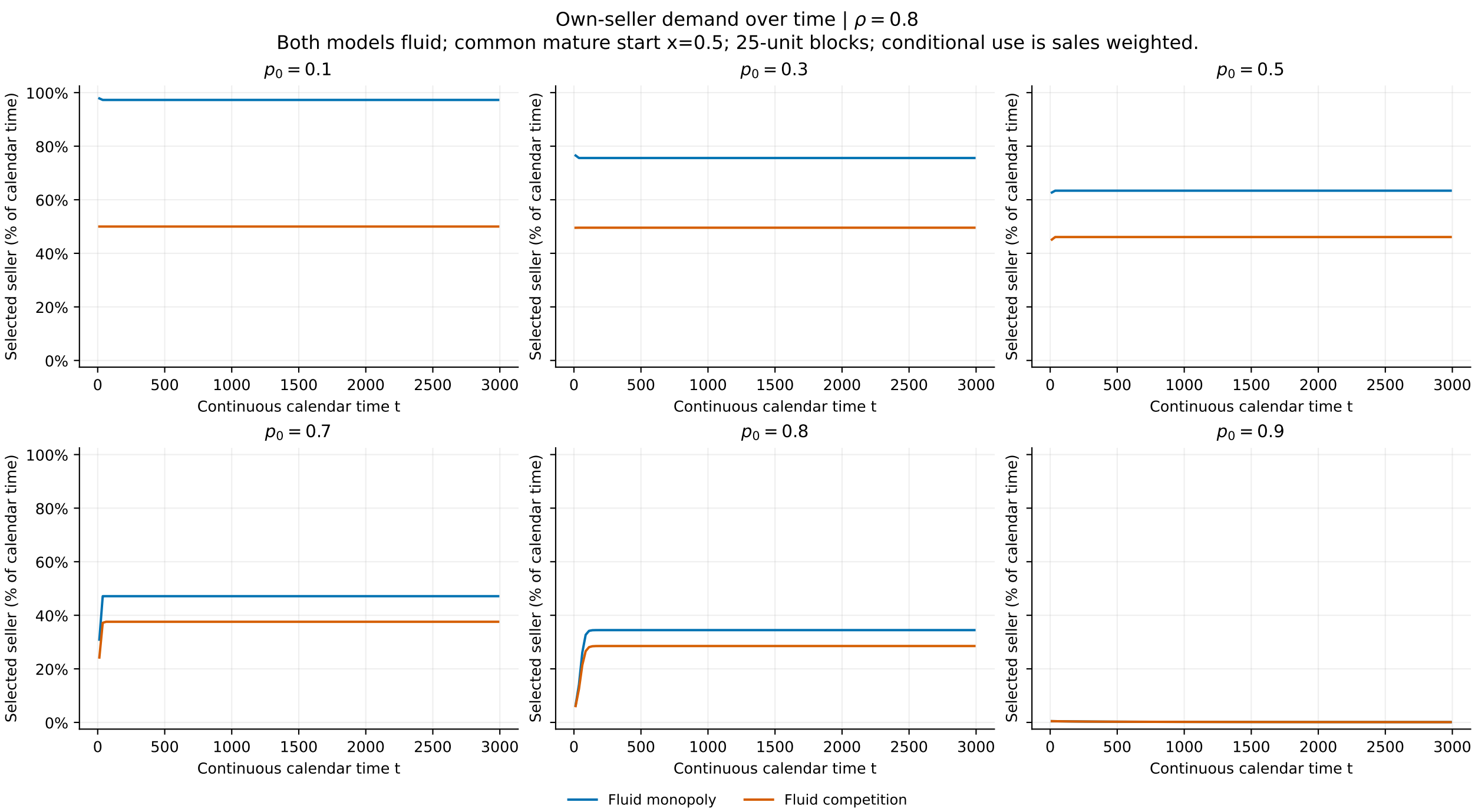

Own-seller demand over time | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

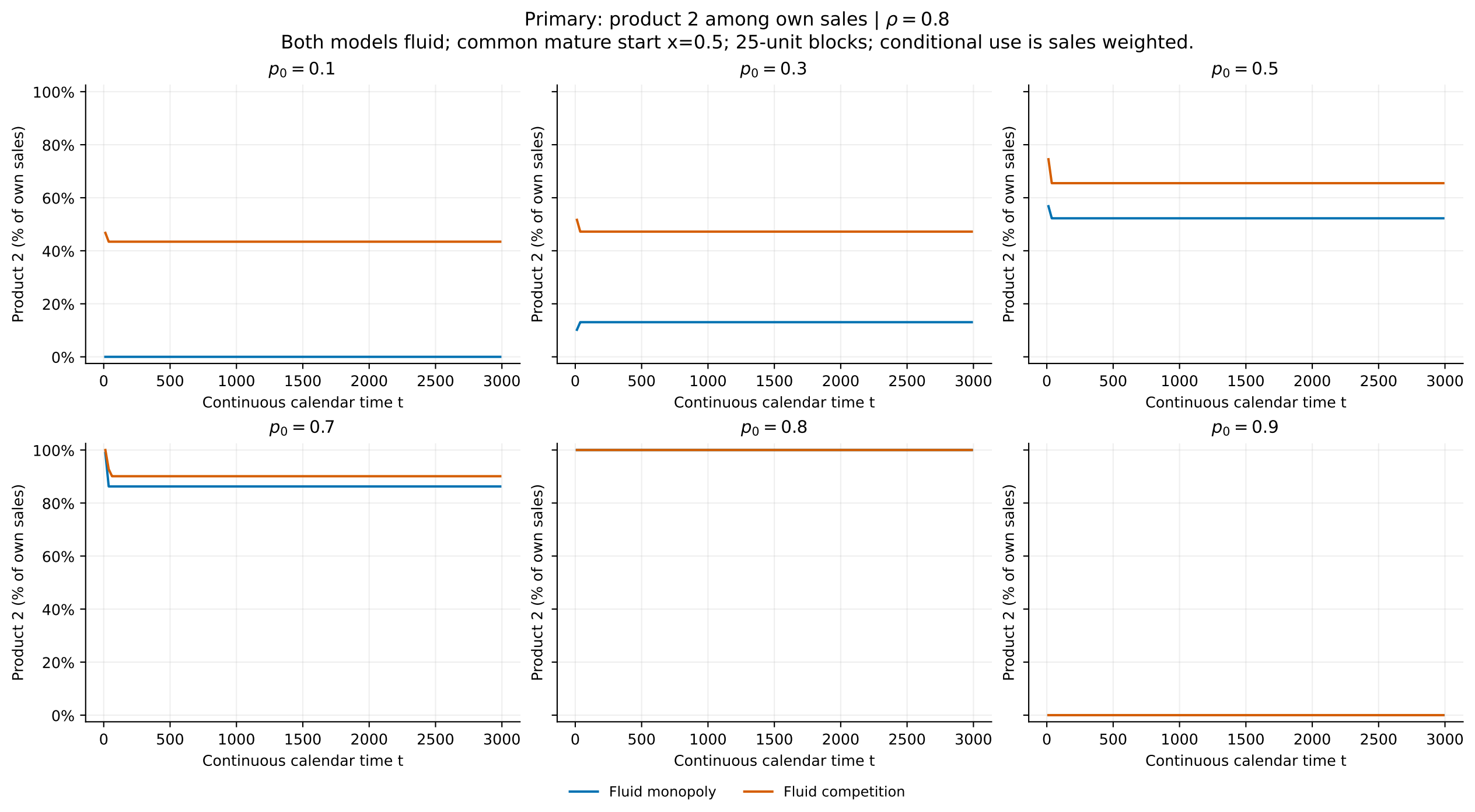

Primary: product 2 among own sales | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

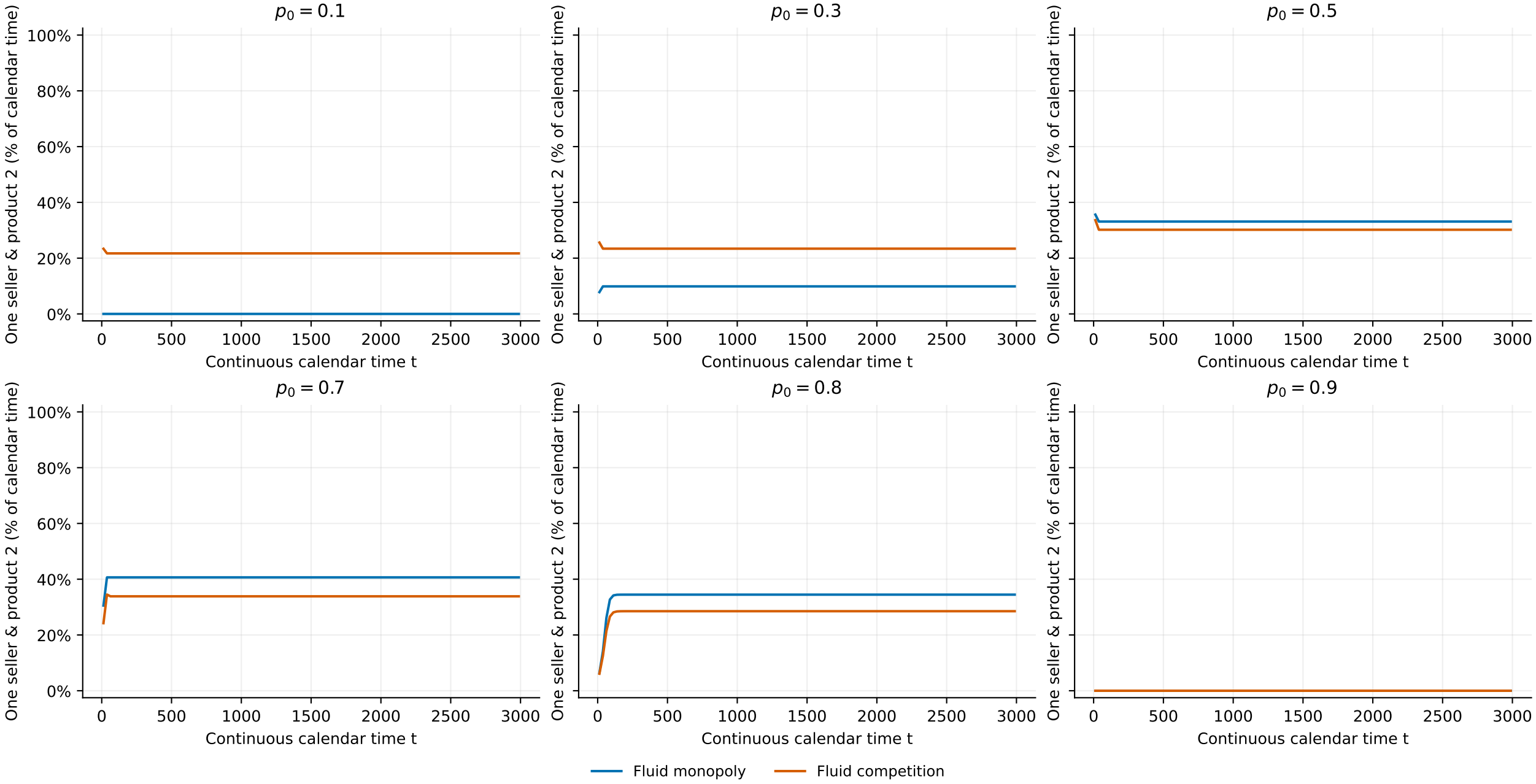

Own seller selected and uses product 2 | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
Continuous calendar time t
0%
20%
40%
60%
80%
100%
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

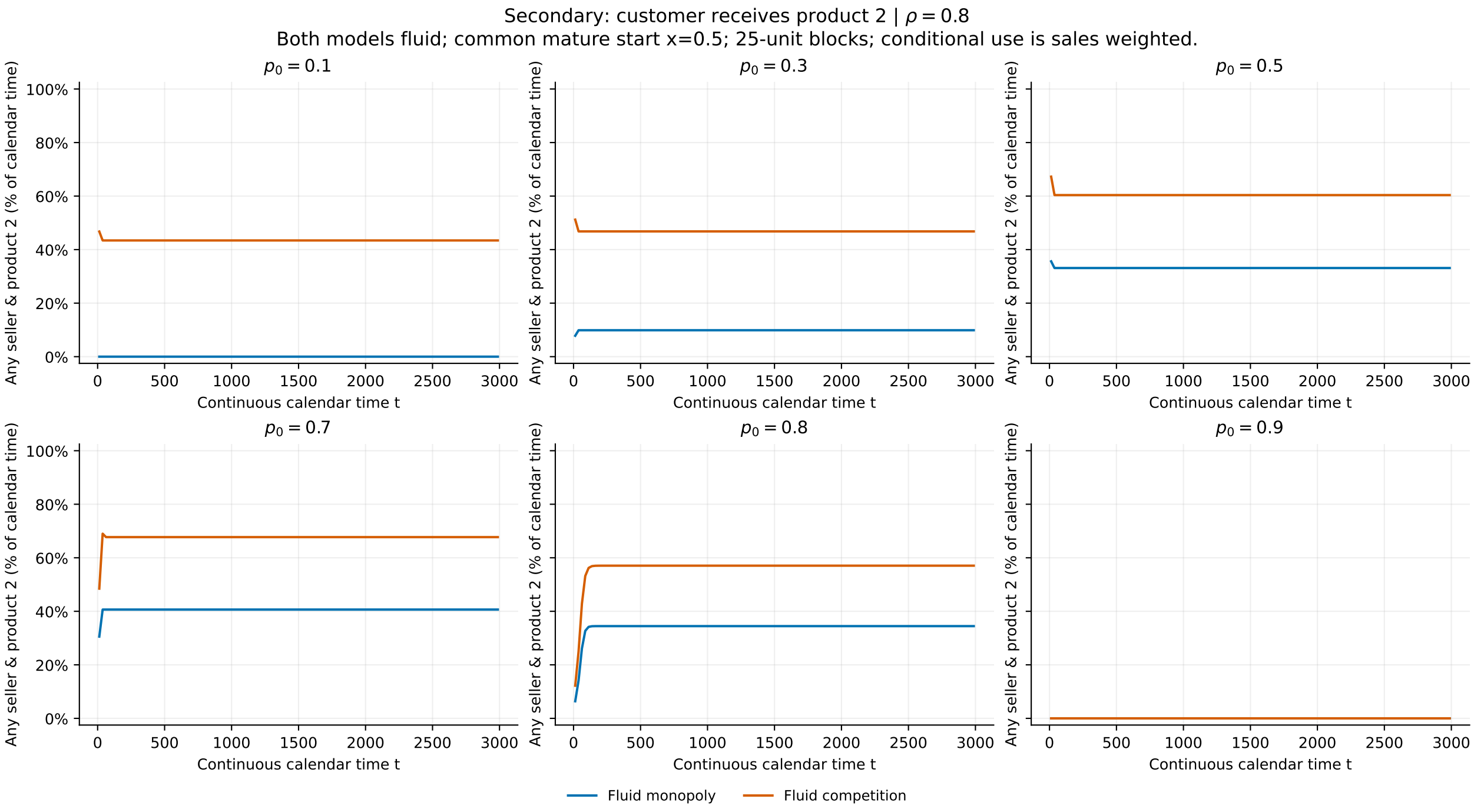

Secondary: customer receives product 2 | $\rho = 0.7$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

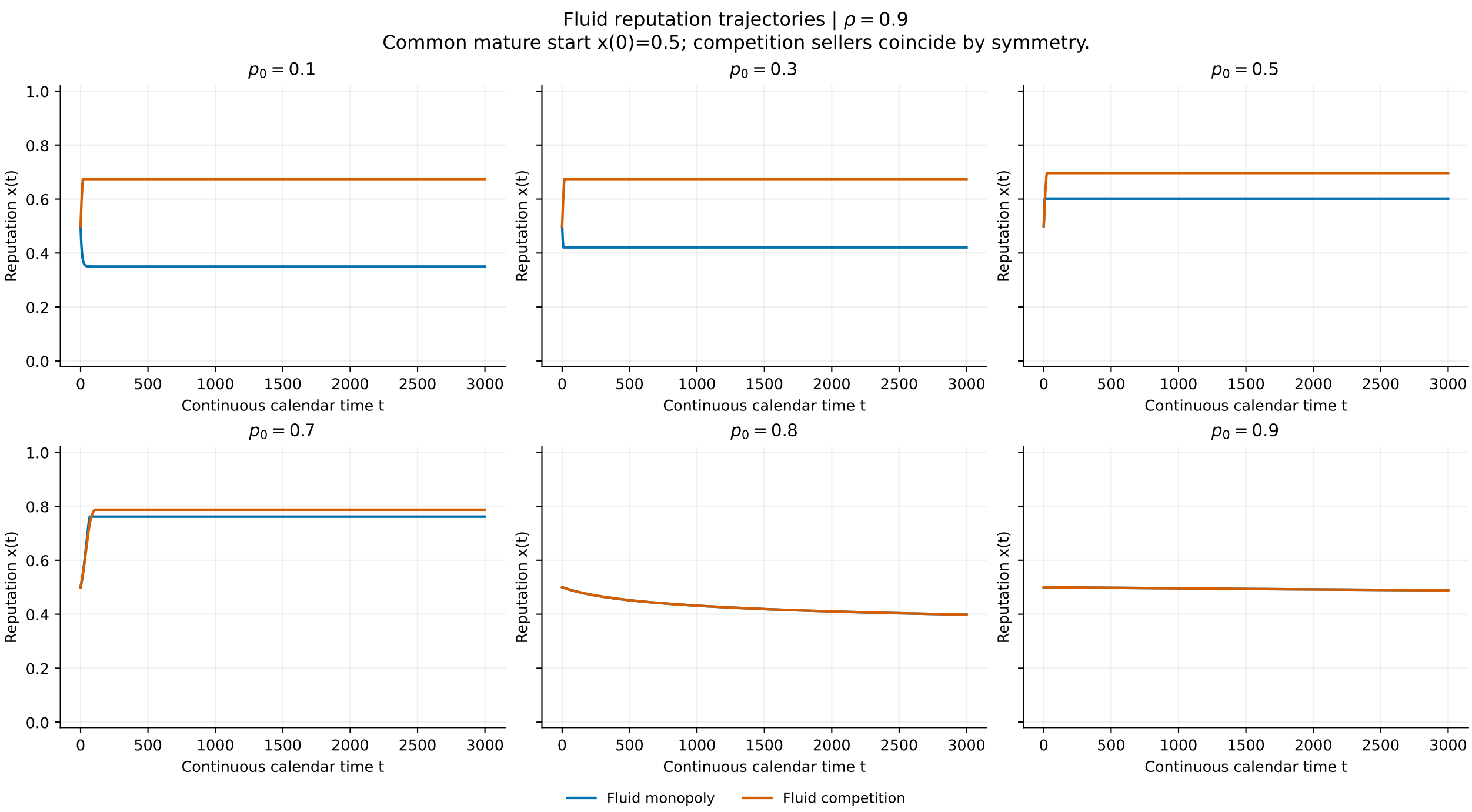

Fluid reputation trajectories | $\rho = 0.8$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition

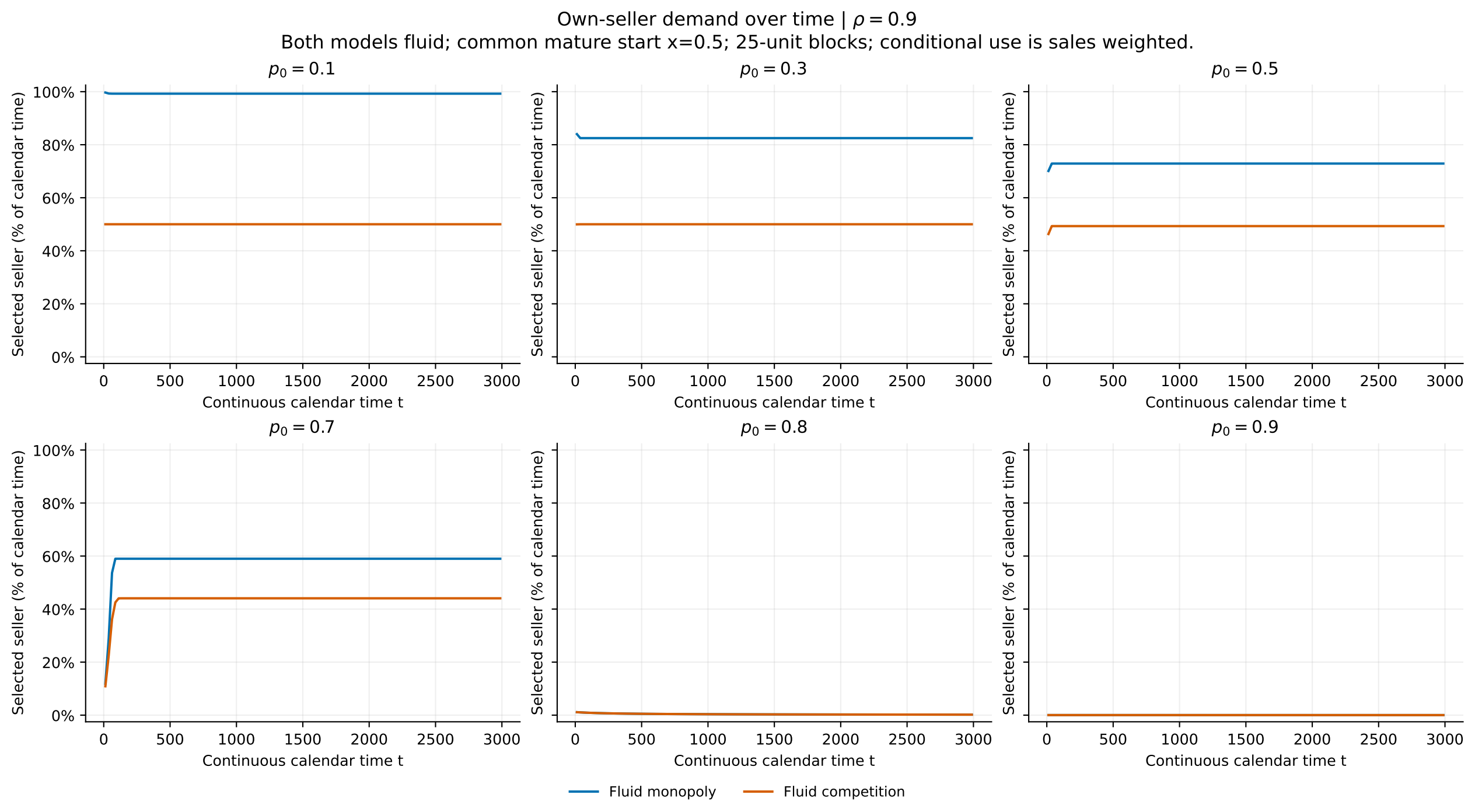

Own-seller demand over time | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

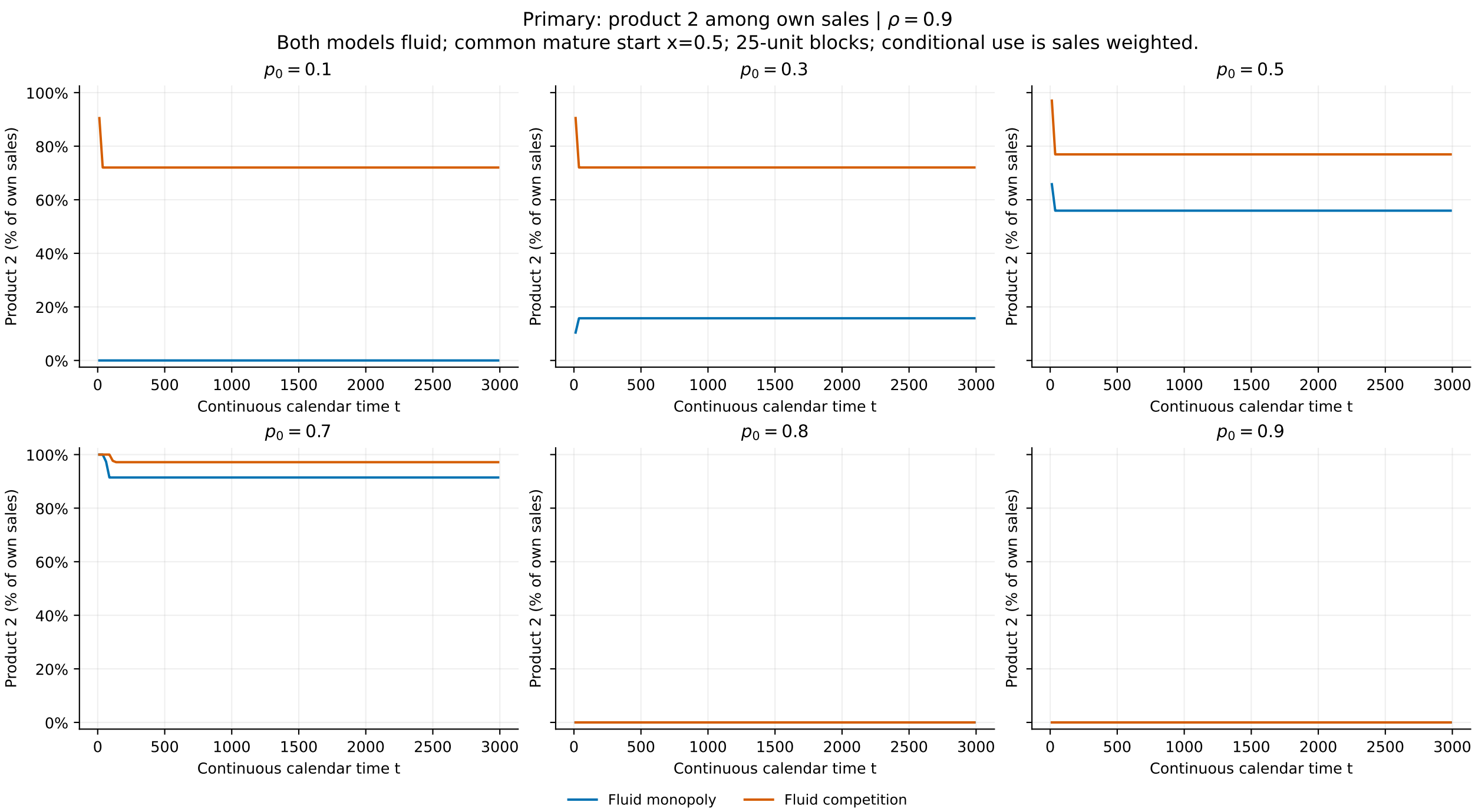

Primary: product 2 among own sales | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

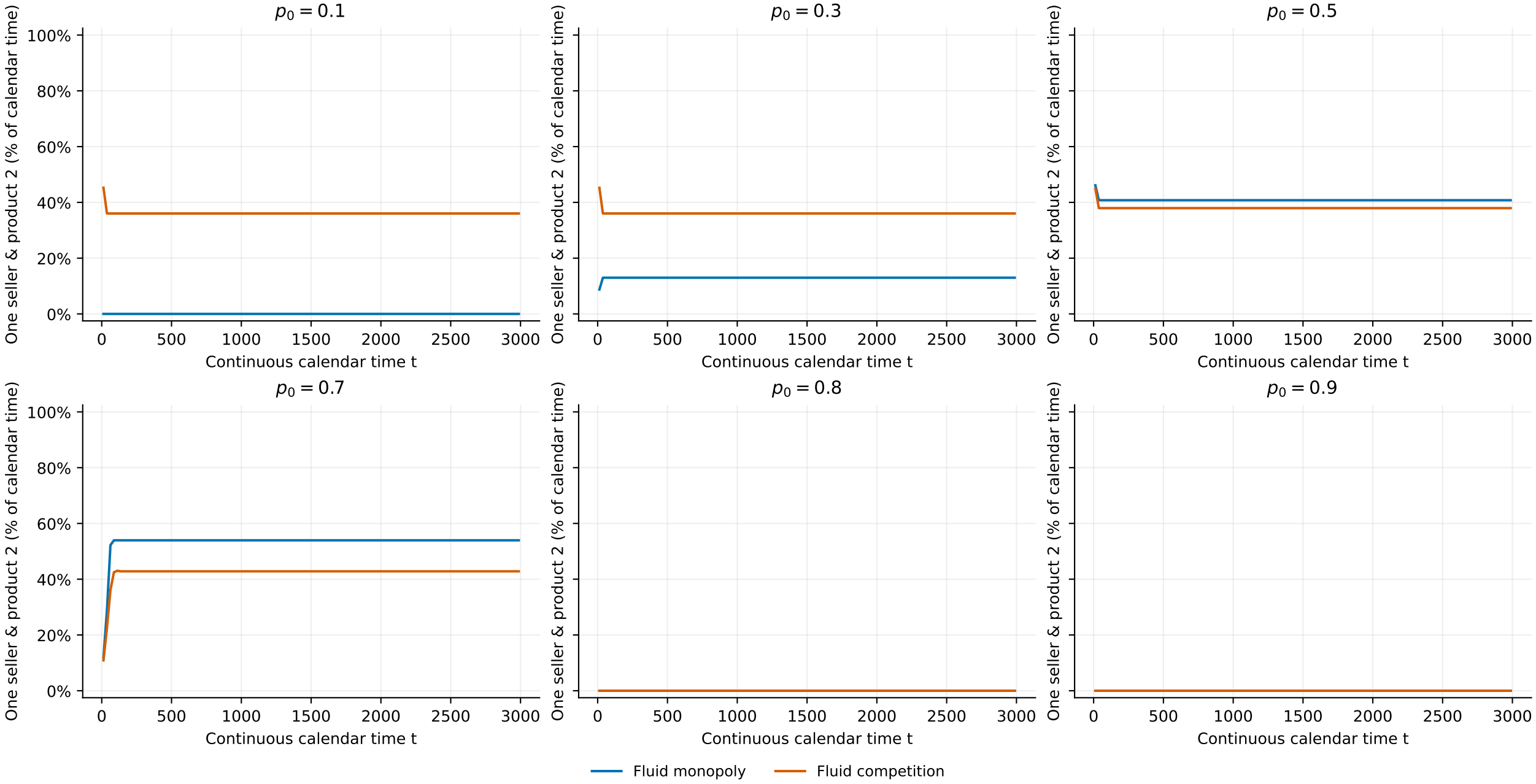

Own seller selected and uses product 2 | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition

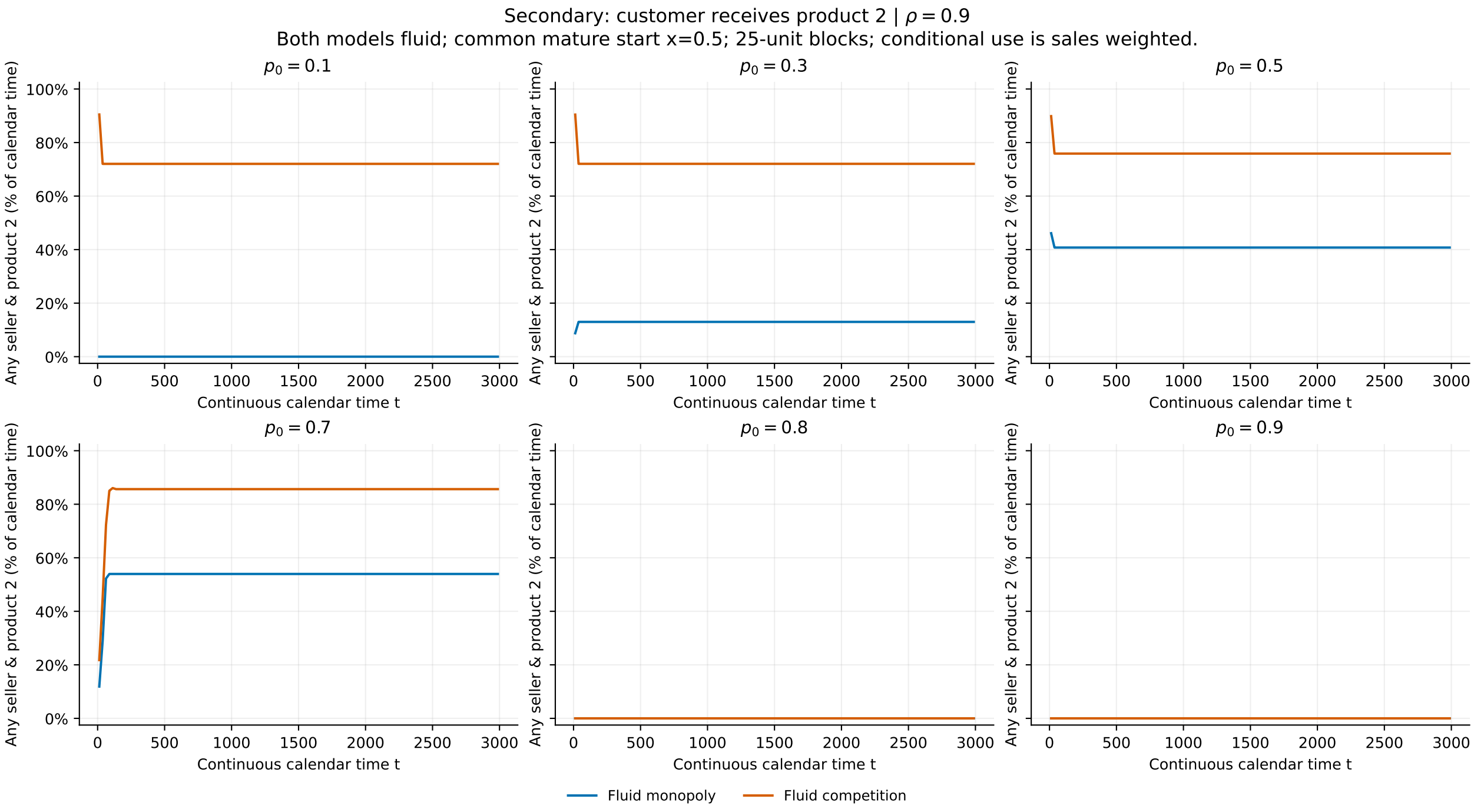

Secondary: customer receives product 2 | $\rho = 0.8$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition


Fluid reputation trajectories | $\rho = 0.9$
Common mature start x(0)=0.5; competition sellers coincide by symmetry.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Reputation x(t)
Continuous calendar time t
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
2500
3000
Fluid monopoly
Fluid competition


Own-seller demand over time | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Selected seller (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition


Primary: product 2 among own sales | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Product 2 (% of own sales)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition


Own seller selected and uses product 2 | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
One seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition


Secondary: customer receives product 2 | $\rho = 0.9$
Both models fluid; common mature start x=0.5; 25-unit blocks; conditional use is sales weighted.
$p_0 = 0.1$
$p_0 = 0.3$
$p_0 = 0.5$
$p_0 = 0.7$
$p_0 = 0.8$
$p_0 = 0.9$
Any seller & product 2 (% of calendar time)
100%
80%
60%
40%
20%
0%
0
500
1000
1500
2000
2500
3000
Continuous calendar time t
Fluid monopoly
Fluid competition